Medtronic

INSTRUCTIONS FOR USE

Solitaire™ X

Revascularization Device

en TABLE OF CONTENTS

Solitaire™ X Revascularization Device

English en
Instructions for Use

Solitaire™ X Revascularization Device

PRECAUTIONS

- The Solitaire™ X Revascularization Device should only be used by physicians trained in interventional neuroradiology and treatment of ischemic stroke.
- Operators should take all necessary precautions to limit X-ray radiation doses to patients and themselves by using sufficient shielding, reducing fluoroscopy times, and modifying X-ray technical factors whenever possible.
- Carefully inspect the sterile package and the Solitaire™ X Revascularization Device prior to use to verify that neither has been damaged during shipment. Do not use kinked or damaged components.
- The Solitaire™ X Revascularization Device is not to be used after the expiration date imprinted on the product label.
- Refer to the appropriate intravenous tissue plasminogen activator (IV t-PA) manufacturer labeling for indications, contraindications, warnings, precautions, and instructions for use.
- Initiate mechanical thrombectomy treatment as soon as possible.
- For indication 3, endovascular therapy with the device should be started within 16 hours of symptom onset.
- For indication 3, users should validate their imaging analysis techniques to ensure robust and consistent results for assessing core infarct size.

DESCRIPTION

The Solitaire™ X Revascularization Device is designed to restore blood flow by removing thrombus in patients experiencing ischemic stroke due to large intracranial vessel occlusion. The device is designed for use in the neurovasculature such as the internal carotid artery, M1 and M2 segments of the middle cerebral artery, basilar, and the vertebral arteries.

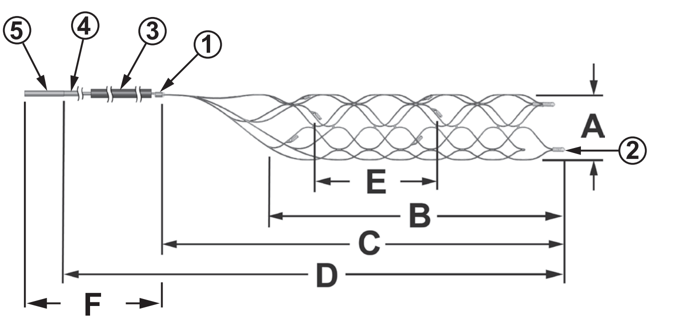


Figure 1.
Solitaire™ X Revascularization Device

1. Proximal Marker
2. Distal Markers
3. Introducer Sheath
4. Fluorosafe Marker
5. Push Wire

A. Stent Diameter
B. Usable Length
C. Stent Length
D. Length from Distal Tip to Fluorosafe Marker
E. Marker to Marker Length
F. Push wire Length

INDICATIONS

1. The Solitaire™ X Revascularization Device is indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA). Endovascular therapy with the device should be started within 6 hours of symptom onset.
2. The Solitaire™ X Revascularization Device is indicated to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset. Patients who are ineligible for IV t-PA or who fail IV t-PA therapy are candidates for treatment.
3. The Solitaire™ X Revascularization Device is indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion of the internal carotid artery (ICA) or middle cerebral artery (MCA)-M1 segments with smaller core infarcts (<70 cc by CTA or MRA, <25 cc by MR-DWI). Endovascular therapy with the device should start within 6-16 hours of time last seen well in patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy.

CONTRAINDICATIONS

Use of the Solitaire™ X Revascularization Device is contraindicated under these circumstances.

- Patients with known hypersensitivity to nickel-titanium.
- Patients with stenosis and/or pre-existing stent proximal to the thrombus site that may preclude safe recovery of the Solitaire™ X Revascularization Device.
- Patients with angiographic evidence of carotid dissection.

POTENTIAL COMPLICATIONS

Possible complications include, but are not limited to the following:

- Adverse reaction to antiplatelet/ anticoagulation agents or contrast media
- Air Embolism
- Allergic reactions
- Angina
- Arrhythmia
- Arteriovenous Fistula
- Aspiration
- Brain Edema
- Cancer
- Change in mental status
- Coagulopathy
- Death
- Device(s) deformation, collapse, fracture or malfunction
- Distal embolization including to a previously uninvolved territory
- Embolism
- Fever
- Fistula
- Foreign body in patient
- Foreign body reaction
- Hemolysis
- Hemorrhage
- Hemorrhagic stroke
- Hypersensitivity
- Hypertension
- Hypotension
- Infarction, cerebral
- Infection
- Inflammation
- Ischemia
- Myocardial Infarction
- Necrosis
- Nerve damage
- Neurologic deterioration including stroke progression, stroke in new vascular territory, and death
- Organ failure
- Pain
- Perforation or dissection of the vessel
- Persistent neurological deficits
- Radiation exposure, unintended
- Rupture
- Shock
- Stenosis
- The risk of complication of radiation exposure (e.g., alopecia, burns ranging in severity from skin reddening to ulcers, cataracts, and delayed neoplasia) increases as the procedure time and the number of procedures increase
- Therapeutic response decreased
- Thromboembolism
- Thrombosis (acute and subacute)
- Toxicity
- User experiences major dissatisfaction with device performance
- Vascular occlusion
- Vasoconstriction (Vasospasm)
- Vision symptoms

WARNINGS - ALL INDICATIONS

- The appropriate anti-platelet and anti-coagulation therapy should be administered in accordance with standard medical practice.
- Administer IV t-PA as soon as possible for all patients who are indicated to receive the drug. Do not cause delays in this therapy.
  - Per IV t-PA manufacturer labeling, IV t-PA should be administered within 3 hours of stroke symptom onset (IV t-PA use beyond 3 hours is not approved in the United States).
- Do not torque the Solitaire™ X Revascularization Device.
- For vessel safety, do not perform more than three recovery attempts in the same vessel using Solitaire™ X Revascularization Devices.
- For device safety, do not use each Solitaire™ X Revascularization Device for more than three flow restoration recoveries.
- For each new Solitaire™ X Revascularization Device, use a new microcatheter.
- Solitaire™ X Revascularization Device does not allow for electrolytic detachment.
- To prevent device separation:
  - Do not oversize device.
  - Do not recover (i.e. pull back) the device when encountering excessive resistance. Instead, resheath the device with the microcatheter and then remove the entire system under aspiration. If resistance is encountered during resheathing, discontinue and remove the entire system under aspiration.
  - Do not treat patients with known stenosis proximal to the thrombus site.
- This device is supplied STERILE for single use only. Do not reprocess or re-sterilize. Reprocessing and re-sterilization increase the risks of patient infection and compromised device performance.

WARNINGS - INDICATION 1 & 3 ONLY

- The safety and effectiveness has not been established for the Solitaire™ X device to reduce disability in patients with the following:
  - Posterior circulation occlusions
  - More distal occlusions in the anterior circulation
  - Large core infarct (ASPECTS ≤7)

CLINICIAN USE INFORMATION

Materials Required

The following parts are required to use the Solitaire™ X Revascularization Device:

- Solitaire™ X Revascularization Devices should be introduced through a microcatheter with an inside diameter of 0.017 inches (0.43 mm)-0.027 inches (0.69 mm). Refer to Table 1 for recommended microcatheter sizing guidelines.

Table 1.
Solitaire™ X Revascularization Device:
Product Specifications and Recommended Sizing Guidelines

Model	Recommended Vessel Diameter[1]	Recommended Microcatheter Inner Diameter	Push Wire Length	Stent Diameter	Usable Length[2]	Stent Length	Length from Distal Tip to Fluorosafe Marker	Distal Radio-paque Markers	Proximal Radio-paque Markers	Radiopaque Stent Markers Spacing
SFR4-3-20-10	1.5 mm-3.0 mm	0.017" (0.43 mm)-0.027" (0.69 mm)	200 cm	3.0 mm	20.0 mm	30.6 mm	<150 cm	3	1	10 mm
SFR4-3-40-10	1.5 mm-3.0 mm	0.017" (0.43 mm)-0.027" (0.69 mm)	200 cm	3.0 mm	40.0 mm	51.6 mm	<150 cm	3	1	10 mm
SFR4-4-20-05	1.5 mm-4.0 mm	0.021" (0.53 mm)-0.027" (0.69 mm)	200 cm	4.0 mm	20.0 mm	31.0 mm	<130 cm	3	1	5 mm
SFR4-4-20-10	1.5 mm-4.0 mm	0.021" (0.53 mm)-0.027" (0.69 mm)	200 cm	4.0 mm	20.0 mm	31.0 mm	<130 cm	3	1	10 mm
SFR4-4-40-10	1.5 mm-4.0 mm	0.021" (0.53 mm)-0.027" (0.69 mm)	200 cm	4.0 mm	40.0 mm	50.0 mm	<130 cm	3	1	10 mm
SFR4-6-20-10	2.0 mm-5.5 mm	0.021" (0.53 mm)-0.027" (0.69 mm)	200 cm	6.0 mm	20.0 mm	31.0 mm	<130 cm	4	1	10 mm
SFR4-6-24-06	2.0 mm-5.5 mm	0.021" (0.53 mm)-0.027" (0.69 mm)	200 cm	6.0 mm	24.0 mm	37.0 mm	<130 cm	4	1	6 mm
SFR4-6-40-10	2.0 mm-5.5 mm	0.021" (0.53 mm)-0.027" (0.69 mm)	200 cm	6.0 mm	40.0 mm	47.0 mm	<130 cm	4	1	10 mm

[1] Select a Solitaire™ X Revascularization Device based on the sizing recommendations in Table 1 and based on the smallest vessel diameter at thrombus site.

[2] Select a Solitaire™ X Revascularization Device usable length that is at least as long as the length of the thrombus.

Compatibility has been established to support all Solitaire™ X Revascularization devices with Phenom™ 21/27, Rebar™ 18, and Marksman™ microcatheters. Additionally, compatibility testing of the 3 mm Solitaire™ X devices has been performed with the Trevo™* Pro-14 microcatheter. Refer to the instructions supplied with all interventional devices and materials to be used in conjunction with the Solitaire™ X Revascularization Device for their intended uses, contraindications and potential complications.

Other accessories for performing a procedure and NOT supplied; to be selected based on the physician's experience and preferences:

- Minimum 5F (1.67 mm) guide catheter
  (**NOTE:** Users should select a guide catheter with appropriate support to deliver interventional devices. For a guide catheter a minimum inside diameter 0.061 inches (1.55 mm) is recommended. For a balloon guide catheter ensure to use 8F (2.67 mm)- 9F (3.00 mm) minimum inside diameter 0.075 inches (1.91 mm))
- Microcatheter (refer to Table 1)
- Guidewire
- Aspiration Device (60 cc syringe or aspiration pump/system, such as Riptide™ Aspiration System)
  NOTE: The Riptide™ Aspiration System has been bench tested with the Solitaire™ Revascularization Device as an aspiration device during retrieval to the guide catheter. Other comparable aspiration pump systems have been reported in clinical literature, although these devices have not been specifically tested with the Solitaire™ Revascularization Devices.
- Saline solution/heparin-saline continuous flush set
- Rotating Hemostasis Valve (RHV)
- Infusion stand
- Femoral arterial lock

PREPARATION AND PROCEDURE

Preparation

1. Administer anti-coagulation and anti-platelet medications per standard institutional guidelines.
2. Aided by angiographic radiography, determine the location and size of the area to be revascularized.
3. Select a Solitaire™ X Revascularization Device per Table 1.
4. To achieve optimal performance of the Solitaire™ X Revascularization Device and to reduce the risk of thromboembolic complications, maintain continuous flushing action between a) the femoral arterial sheath and the guide catheter, b) the microcatheter and the guide catheter and c) the microcatheter and the push wire and the Solitaire™ X Revascularization Device. Check all connections to make sure that during the continuous flush no air enters the guide catheter or the microcatheter.
5. Position a suitable guide catheter as close to thrombus site as possible employing a standard method. The guide catheter should be appropriately sized to retrieve clot if so desired in subsequent steps. Connect a RHV to the fitting of the guide catheter, and then connect a tube to the continuous flush.
6. With the aid of Table 1, select a microcatheter suitable for advancing the Solitaire™ X Revascularization Device.
7. Connect a second RHV to the fitting of the microcatheter and then connect a tube to the continuous flush.
8. Set the flush rate per standard institutional guidelines.
9. With the aid of a suitable guide wire, advance the microcatheter until the end of the microcatheter is positioned distal to the thrombus so that the usable length portion of the Solitaire™ X Revascularization Device will extend past each side of the thrombus in the vessel when fully deployed. Tighten the RHV around the microcatheter. Remove guide wire.

Delivering the Solitaire™ X Revascularization Device

10. Insert the distal end of the introducer sheath partially into the RHV connected to the microcatheter. Tighten the RHV and verify that fluid exits the proximal end of the introducer sheath.
11. Loosen the RHV and advance the introducer sheath until it is firmly seated in the hub of the microcatheter. Tighten the RHV around the introducer sheath to prevent back flow of blood, but not so tight as to damage the Solitaire™ X Revascularization Device during its introduction into the microcatheter. Confirm that there are no air bubbles trapped anywhere in the system.
12. Transfer the Solitaire™ X Revascularization Device into the microcatheter by advancing the push wire in a smooth, continuous manner. Once the flexible portion of the push wire has entered the microcatheter shaft, loosen the RHV and remove the introducer sheath over the proximal end of the push wire. Once completed, tighten the RHV around the push wire. Leaving the introducer sheath in place will interrupt normal infusion of flushing solution and allow back flow of blood into the microcatheter.
13. Visually verify that the flushing solution is infusing normally. Once confirmed, loosen the RHV to advance the push wire.
14. Once the Solitaire™ X Revascularization Device has been advanced to the fluorosafe marker band begin fluoroscopic imaging. With the aid of fluoroscopic monitoring, carefully advance the Solitaire™ X Revascularization Device until its distal markers line up at the end of the microcatheter. The Solitaire™ X Revascularization Device should be positioned such that the usable length portion of the device will extend past each side of the thrombus in the vessel when the device is fully deployed.

WARNING - ALL INDICATIONS

- If excessive resistance is encountered during the delivery of the Solitaire™ X Revascularization Device, discontinue the delivery and identify the cause of the resistance. Advancement of the Solitaire™ X Revascularization Device against resistance may result in device damage and/or patient injury.

Deploying the Solitaire™ X Revascularization Device

15. Loosen the RHV around the microcatheter. To deploy the Solitaire™ X Revascularization Device, fix the push wire to maintain the position of the device while carefully withdrawing the microcatheter in the proximal direction.

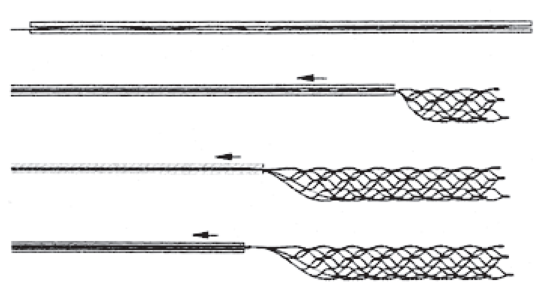


Figure 2.
Solitaire™ X Revascularization Device Deployment

16. Retract the microcatheter until it is just proximal to the proximal marker of the Solitaire™ X Revascularization Device. Tighten the RHV to prevent any movement of the push wire. The usable length of the deployed Solitaire™ device should extend past each side of the thrombus.
17. Tighten the RHV around the microcatheter. Angiographically assess the revascularization status of the treated vessel.

Solitaire™ X Revascularization Device Re-Sheathing

If resheathing of the Solitaire™ X Revascularization Device is necessary (e.g. repositioning), follow these steps:

WARNINGS - ALL INDICATIONS

- Advancing the microcatheter while the device is engaged in clot may lead to embolization of debris.
- Do not advance the microcatheter against any resistance.
- Do not reposition more than two times.

18. Loosen the RHV around the microcatheter and around the push wire. With the aid of fluoroscopic monitoring, hold the push wire firmly in its position to prevent the Solitaire™ X Revascularization Device from moving.
19. Carefully re-sheath the Solitaire™ X Revascularization Device by advancing the microcatheter over the Solitaire™ X Revascularization Device until the distal markers of the Solitaire™ X Revascularization Device line up at the end of the microcatheter as illustrated in Figure 3 below. **If significant resistance is encountered during the re-sheathing process, stop immediately** and proceed to the section below entitled "Revascularization Device Recovery".

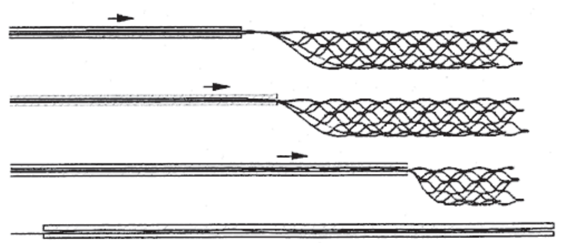


Figure 3.
Solitaire™ X Revascularization Device Re-sheathing

Revascularization Device Recovery

20. If using balloon guide catheter, inflate guide catheter balloon to occlude vessel as specified in Balloon Guide Catheter labeling.
21. Loosen the RHV around the microcatheter enough for movement while still maintaining a seal. If desired remove the microcatheter completely.
22. To retrieve thrombus, withdraw the Solitaire™ X Revascularization Device to the guide catheter tip while applying aspiration to the guide catheter with the aspiration device. If applicable withdraw the microcatheter and the Solitaire™ X Revascularization Device as a unit. Never advance the deployed Solitaire™ X Revascularization Device distally.
23. Apply aspiration to the guide catheter using the aspiration device and recover the Solitaire™ X Revascularization Device inside guide catheter. Continue aspirating guide catheter until the Solitaire™ X Revascularization Device is nearly withdrawn from the guide catheter.
    NOTE: If withdrawal into the guide catheter is difficult, deflate balloon (if using balloon guide catheter) and then simultaneously withdraw guide catheter, microcatheter and Solitaire™ X Revascularization Device as a unit through the sheath while maintaining aspiration. Remove sheath if necessary.

WARNINGS - ALL INDICATIONS

- If excessive resistance is encountered during recovery of the Solitaire™ X Revascularization Device, discontinue the recovery and identify the cause of the resistance.
- For vessel safety, do not perform more than three recovery attempts in the same vessel using the Solitaire™ X Revascularization Device.

24. Open the guide catheter RHV to allow the Solitaire™ X Revascularization Device to exit without resistance. If applicable allow the microcatheter and the Solitaire™ X Revascularization Device to exit without resistance. Use care to avoid interaction with the site of the intervention and to prevent air from entering the system.
25. Aspirate the guide catheter to ensure the guide catheter is clear of any thrombus material.
26. Deflate guide catheter balloon if using balloon guide catheter.
27. If additional flow restoration attempts are desired with:
    a. **A new Solitaire™ X Revascularization Device, then:**
        i. Repeat the steps described above starting with the "Preparation" section.
    b. **The same Solitaire™ X Revascularization Device, then:**
        i. Clean the device with saline solution.
        NOTE: Do not use solvents or autoclave.
        ii. Carefully inspect the device for damage.
        - If no damage, load the introducer sheath over the proximal end of the push wire. For 3 mm devices, one end of the introducer sheath is tapered and should be (1) aligned with the distal end of the Solitaire™ X Revascularization Device and (2) interface with the microcatheter hub.
        - If there is any damage, do not use the device and use a new Solitaire™ X Revascularization Device for subsequent flow restoration attempts following the steps described above starting with the "Preparation" section. Use of a damaged device could result in additional device damage or patient injury.

WARNING - ALL INDICATIONS

- For device safety, do not use each Solitaire™ X Revascularization Device for more than three flow restoration recoveries.

HOW SUPPLIED

The Solitaire™ X Revascularization Device is provided sterile for single patient use only.

Sterile: This device is sterilized with Ethylene Oxide. Non-pyrogenic.

Contents: One (1) Solitaire™ X Revascularization Device.

Storage: Store product in a dry, cool place.

SAFETY AND EFFECTIVENESS INFORMATION – INDICATION 1

SWIFT PRIME Clinical Study

SWIFT PRIME (Solitaire™ FR or Solitaire™ 2 With the Intention For Thrombectomy as Primary Endovascular Treatment for Acute Ischemic Stroke) is a global, multicenter, two-arm, prospective, randomized, open, blinded endpoint (PROBE) clinical study comparing neurological disability outcomes (defined by mRS) in Acute Ischemic Stroke (AIS) patients who are treated with either IV t-PA alone or IV t-PA in combination with Solitaire™ FR or Solitaire™ 2 mechanical thrombectomy intervention. Subjects receiving IV t-PA within 4.5 hours of symptom onset were randomized 1:1 to mechanical thrombectomy with Solitaire™ within 6 hours of onset, or to continuation with IV t-PA alone. Within this group, the Analysis cohort was defined as those subjects who were administered IV t-PA within 3 hours of symptom onset.

Inclusion Criteria

Subjects were considered eligible for the study if they satisfied all of the following inclusion criteria:

- Subject or subject's legally authorized representative has signed and dated an Informed Consent Form
- Age 18 – 80
- Clinical signs consistent with acute ischemic stroke
- Pre-stroke Modified Rankin Score less than or equal to 1
- National Institute of Health Stoke Scale (NIHSS) score of at least 8 and less than 30 at the time of randomization
- Initiation of IV t-PA within 4.5 hours of onset of stroke symptoms
- Thrombolysis in Cerebral Infarction (TICI) 0 to 1 flow in the intracranial internal carotid artery, M1 segment of the MCA, or carotid terminus confirmed by CT or MR angiography that is accessible to the Solitaire™ FR or Solitaire™ 2 device
- Subject is able to be treated within 6 hours of stroke symptoms onset and within 1.5 hours from CTA or MRA to groin puncture
- Subject is willing to conduct follow-up visits

Exclusion Criteria

Subjects were considered ineligible for study participation if they met any of the following exclusion criteria:

- Subject who is contraindicated to IV t-PA as per local national guidelines
- Females who are pregnant or lactating
- Rapid neurological improvement prior to randomization suggesting resolution of signs/symptoms of stroke
- Known serious sensitivity to radiographic contrast agents
- Known sensitivity to Nickel, Titanium metals or their alloys
- Current participation in another investigational drug or device study
- Known hereditary or acquired hemorrhagic diathesis, coagulation factor deficiency
- Renal failure as defined by a serum creatinine greater than 2.0 mg/dl (or 176.8 µmol/l) or Glomerular Filtration Rate [GFR] less than 30
- Requires hemodialysis or peritoneal dialysis, or who have contraindication to an angiogram
- Life expectancy less than 90 days
- Clinical presentation suggests a subarachnoid hemorrhage, even if initial CT or MRI scan is normal
- Suspicion of aortic dissection
- Co-morbid disease or condition that would confound the neurological and functional evaluations or compromise survival or ability to complete follow-up assessments
- Currently uses or has a recent history of illicit drug(s) or abuses alcohol
- Known history of arterial tortuosity, pre-existing stent, and/or other arterial disease which would prevent the device from reaching the target vessel and/or preclude safe recovery of the device
- Additionally, subjects were also considered ineligible for study participation if they met any of the following imaging exclusion criteria:
    - CT or MRI evidence of hemorrhage on presentation, mass effect or intracranial tumor (except small meningioma), cerebral vasculitis, basilar artery (BA) occlusion or posterior cerebral artery (PCA) occlusion, carotid dissection, or complete cervical carotid occlusion requiring stenting at the time of the index procedure
    - CT showing hypodensity or MRI showing hyperintensity involving greater than 1/3 of the middle cerebral artery (MCA) territory (or in other territories, >100 cc of tissue) on presentation
    - Baseline non-contrast CT or DWI MRI evidence of a moderate/large core defined as extensive early ischemic changes of Alberta Stroke Program Early CT score (ASPECTS) less than 6. Patients enrolled under RAPID were excluded based on the following:
        a. MRI- or CT-assessed core infarct lesion greater than 50 cc; or
        b. Severe hypoperfusion lesion (10 sec or more Tmax lesion larger than 100 cc; or
        c. Ischemic penumbra < 15 cc and mismatch ratio ≤1.8.
    - Evidence that suggests, in the opinion of the investigator, the subject is not appropriate for mechanical thrombectomy intervention

Study Endpoints

The primary effectiveness endpoint of the study is 90-day global disability assessed via the blinded evaluation of modified Rankin Scale (mRS). Secondary clinical efficacy endpoints of the study are:

- Death due to any cause at 90 days.
- Functional independence as defined by mRS score ≤ 2 at 90 days.
- Change in NIHSS score at 27 ± 6 hours post randomization.

The secondary technical efficacy endpoints of the study are:

- Volume of cerebral infarction as measured by a CT or MRI scan at 27 ± 6 hours post randomization.
- Reperfusion measured by reperfusion ratio on CT or MRI scan 27 ± 6 hours post randomization
- Arterial revascularization measured by TICI 2b or 3 following device use.
- Correlation of RAPID-assessed core infarct volume with 27 ± 6 hours post randomization stroke infarction in subjects who achieved TICI 2b-3 reperfusion without intracranial hemorrhage.

Results

A total of 196 subjects were randomized into the SWIFT PRIME Study (98 in each group). The SWIFT PRIME study allowed IV t-PA use beyond 3 hours, although IV t-PA is not approved in the United States beyond 3 hours. Patients treated with IV-tPA beyond 3 hours did not factor strongly in the evaluation of the Solitaire™ 2 revascularization device and have been excluded from the analyses. The resulting Analysis Cohort consists of 161 subjects (84 in the IV t-PA plus Solitaire™ group and 77 with IV t-PA only). Additionally, 17 subjects in the IV t-PA plus Solitaire™ group were excluded from the primary and secondary efficacy endpoint analyses. These 17 subjects either received carotid stenting and/or angioplasty or were treated in a manner inconsistent with the Solitaire™ Instructions for Use. Therefore, the primary and secondary efficacy endpoint analyses cohort consists of 144 subjects.

The proportion of patients functionally independent (mRS 0-2) at the 90-day visit was higher in the IV t-PA plus Solitaire™ device group. A summary of the subject disposition is presented below.

Table 2. Summary of Subject Disposition (Analysis Cohort)

Subject Disposition	IV t-PA Only	IV t-PA + Solitaire™	Overall
Attended 90 day visit	63/77 (81.8%)	60/67 (89.6%)	123/144 (85.4%)
Died prior to 90 day visit	10/77 (13.0%)	6/67 (9.0%)	16/144(11.1%)
Early Terminated			
Subject Lost to Follow-Up	3/77 (3.9%)	0/67 (0.0%)	3/144 (2.1%)
Subject Withdrew Consent	1/77 (1.3%)	1/67 (1.5%)	2/144 (1.4%)
Investigator Withdrew the Subject	0/77 (0.0%)	0/67 (0.0%)	0/144 (0.0%)
Total	77 (100.0%)	67 (100.0%)	144 (100.0%)

Standard summary statistics were calculated for all study variables and subject data were analyzed according to the group to which they were randomized. For continuous variables, statistics included means, standard deviations, medians and ranges. Categorical variables were summarized in frequency distributions.

For the primary efficacy endpoint, statistical significance was declared using bounds predefined in the group sequential analysis plan, which accounts for multiplicity due to interim analyses. Elsewhere, one-sided statistical tests having p-values less than 0.025 were deemed significant while two-sided tests having p-values less than 0.05 were deemed significant.

For adverse event reporting, the primary analysis is based on subject counts, not event counts. Both subject counts and event counts are presented in tabular summaries of results.

NOTE: The SFR4-3-20-10, SFR4-3-40-10, SFR4-4-20-05, SFR4-4-20-10, SFR4-4-40-10, SFR4-6-20-10, SFR4-6-24-06 and SFR4-6-40-10 models were not evaluated as part of the SWIFT PRIME study.

Table 3. Primary Effectiveness Endpoint (Analysis Cohort)

mRS Score at 90 days	IV t-PA Only*	IV t-PA + Solitaire™	p-value
			0.0007
0	7.9% (6/76)	16.4% (11/67)	
1	11.8% (9/76)	31.3% (21/67)	
2	17.1% (13/76)	14.9% (10/67)	
3	18.4% (14/76)	11.9% (8/67)	
4	18.4% (14/76)	13.4% (9/67)	
5/6	26.3% (20/76)	11.9% (8/67)	

Missing data at 90-day was imputed using LOCF (except baseline data not used)

*** One subject from the IV t-PA only group withdrew consent 27 hours post randomization. No 90 day mRS Score was available for this subject.**

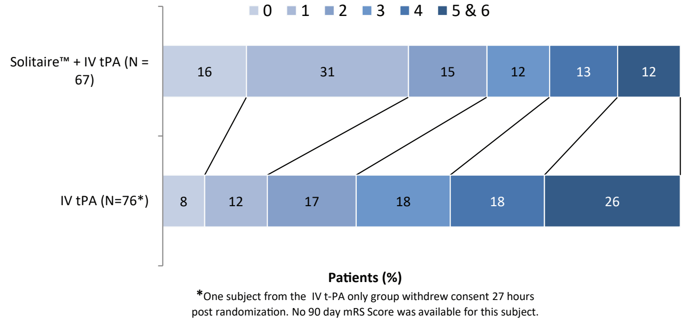


Figure 4. Modified Rankin Shift at 90 days (Analysis Cohort)

Table 4. Secondary Clinical Efficacy Endpoints (Analysis Cohort)

Secondary Clinical Efficacy Endpoint	IV t-PA Only*	IV t-PA + Solitaire™	Odds Ratio/ Difference [95% CI]	p-value
Mortality at 90 days	10/76 (13.2%)	6/67 (9.0%)	0.65 (0.22-1.89)	0.596
Functional Independence (mRS 0-2) at 90 days	28/76 (36.8%)	42/67 (62.7%)	2.88 (1.46-5.68)	0.003**
Change in NIHSS at 27 hours Post randomization				
N	76	66		<0.001**
Mean ± SD	-4.3 ± 6.4	-9.9 ± 7.2	-5.6 (-7.9, -3.3)	
Median	-3.0	-9.5		
(min, max)	(-24.0, 9.0)	(-27.0,10.0)		

*** One subject from the IV t-PA only group withdrew consent 27 hours post randomization. No secondary clinical efficacy endpoints were available for this subject.**

**** These p-values are for informative purposes only. Based on the pre-defined hierarchical testing of the secondary endpoints, these particular endpoints were not achieved.**

Table 5. Secondary Technical Efficacy Endpoints (Analysis Cohort)

Secondary Technical Efficacy Endpoint	IV t-PA Only	IV t-PA + Solitaire™	Difference [95% CI]
Infarct Volume at 27 hours Post-Randomization (cc)**			
N	77	66	
Mean ± SD	68.9 ± 75.5	51.1 ± 86.1	N/A*
Median	46.5	19.2	
(min, max)	(0.0, 406.6)	(0.0,530.5)	
Reperfusion Ratio at 27 hours Post-Randomization**			
N	44	45	
Mean ± SD	61.6 ± 56.1	87.9 ± 37.2	N/A*
Median	84.0	100.0	
(min, max)	(-200.0, 100.0)	(-94.0,100.0)	
TICI 2b-3 Following Device Use**			
% (n/N)	N/A	90.2% (55/61)	N/A

*** Wilcoxon rank-sum test due to non-normality of data.**

**** Per Core Laboratory assessed data**

Table 6. Primary Safety Endpoints (Analysis Cohort)			
Safety Endpoint	**IV t-PA Only**	**IV t-PA + Solitaire™**	**Odds Ratio**
Primary Safety Variables			
Any serious adverse event*	33.8% (26/77)	31.0% (26/84)	0.88 (0.45-1.70)
Symptomatic ICH at 27 hours**	3.9% (3/77)	0.0% (0/84)	NA

NA denotes not applicable

*** Per Clinical Events Committee adjudication**

**** Per Core Laboratory assessed data**

Table 7. Safety Endpoints, RAPID Imaging Requirement Subgroup (Analysis Cohort)			
Safety Endpoint	**IV t-PA Only**	**IV t-PA + Solitaire™**	**Odds Ratio**
All Serious Adverse Events*	43.3% (13/30)	42.9% (12/28)	0.98 (0.35-2.77)
Symptomatic ICH at 27 hours**	0.0% (0/30)	0.0% (0/28)	NA

NA denotes not applicable

*** Per Clinical Events Committee adjudication**

**** Per Core Laboratory assessed data**

Patients enrolled under RAPID were excluded based on the following:

a) MRI- or CT-assessed core infarct lesion greater than 50 cc; or

b) Severe hypoperfusion lesion (10 sec or more Tmax lesion larger than 100 cc; or

c) Ischemic penumbra < 15 cc and mismatch ratio ≤1.8.

Table 8. Safety Endpoints, ASPECTS Imaging Requirement Subgroup (Analysis Cohort)			
Safety Endpoint	**IV t-PA Only**	**IV t-PA + Solitaire™**	**Odds Ratio**
All Serious Adverse Events*	29.8% (14/47)	25.0% (14/56)	0.79 (0.33-1.88)
Symptomatic ICH at 27 hours**	6.4% (3/47)	0.0% (0/56)	NA

NA denotes not applicable

*** Per Clinical Events Committee adjudication**

**** Per Core Laboratory assessed data**

Subjects enrolled under ASPECTS were excluded based on ASPECTS score <6.

Table 9. CEC Adjudicated Adverse Events (Analysis Cohort)			
MedDRA Preferred Term	**IV t-PA Only**	**IV t-PA + Solitaire™**	**All Enrolled**
Units	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]
TOTAL Adverse Events (AE)	**93.5% (72/77) [416]**	**91.7% (77/84) [449]**	**92.5% (149/161) [865]**
Blood and Lymphatic System Disorders	**6.5% (5/77) [5]**	**13.1% (11/84) [11]**	**9.9% (16/161) [16]**
Anaemia	3.9% (3/77) [3]	6.0% (5/84) [5]	5.0% (8/161) [8]
Haemorrhagic Anaemia	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Leukocytosis	1.3% (1/77) [1]	7.1% (6/84) [6]	4.3% (7/161) [7]
Cardiac Disorders	**26.0% (20/77) [23]**	**31.0% (26/84) [31]**	**28.6% (46/161) [54]**
Atrial Fibrillation	3.9% (3/77) [3]	6.0% (5/84) [5]	5.0% (8/161) [8]
Atrial Flutter	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Atrial Thrombosis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Bradycardia	7.8% (6/77) [6]	4.8% (4/84) [4]	6.2% (10/161) [10]
Cardiac Arrest	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Cardiac Failure	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Cardiac Failure Congestive	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Intracardiac thrombus	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Ischaemic cardiomyopathy	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Myocardial infarction	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Palpitations	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Sick sinus syndrome	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Sinus arrest	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Supraventricular tachycardia	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Tachycardia	11.7% (9/77) [9]	8.3% (7/84) [7]	9.9% (16/161) [16]
Ventricular tachycardia	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]
Ear and Labyrinth Disorders	**1.3% (1/77) [1]**	**0.0% (0/84) [0]**	**0.6% (1/161) [1]**
Vertigo	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Endocrine disorders	**0.0% (0/77) [0]**	**1.2% (1/84) [1]**	**0.6% (1/161) [1]**
Hypothyroidism	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Eye disorders	**1.3% (1/77) [1]**	**4.8% (4/84) [4]**	**3.1% (5/161) [5]**
Conjunctival haemorrhage	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Eye haemorrhage	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Optic neuropathy	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Photopsia	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Visual impairment	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Gastrointestinal Disorders	**37.7% (29/77) [40]**	**33.3% (28/84) [43]**	**35.4% (57/161) [83]**
Abdominal pain	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Abdominal pain lower	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Abdominal pain upper	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Constipation	15.6% (12/77) [13]	15.5% (13/84) [13]	15.5% (25/161) [26]
Diarrhea	3.9% (3/77) [3]	1.2% (1/84) [1]	2.5% (4/161) [4]
Dyspepsia	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Dysphagia	9.1% (7/77) [7]	7.1% (6/84) [6]	8.1% (13/161) [13]
Faecal incontinence	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Gastritis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Gastrooesophageal reflux disease	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Haematemesis	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]
Haemorrhoidal haemorrhage	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Ileus	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Intestinal ischaemia	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Lower gastrointestinal haemorrhage	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Melaena	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Mouth haemorrhage	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Nausea	5.2% (4/77) [4]	8.3% (7/84) [7]	6.8% (11/161) [11]
Oesophageal stenosis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Oral pain	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Retroperitoneal haematoma	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Salivary hypersecretion	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Vomiting	3.9% (3/77) [3]	4.8% (4/84) [4]	4.3% (7/161) [7]
General Disorders and Administration Site Conditions	**18.2% (14/77) [16]**	**25.0% (21/84) [24]**	**21.7% (35/161) [40]**
Application site pruritus	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Catheter site haematoma	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Catheter site haemorrhage	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]

Catheter site induration	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Catheter site inflammation	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Cyst	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Discomfort	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Extravasation	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Fatigue	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Influenza like illness	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Local swelling	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Oedema peripheral	5.2% (4/77) [4]	0.0% (0/84) [0]	2.5% (4/161) [4]
Pain	2.6% (2/77) [2]	4.8% (4/84) [4]	3.7% (6/161) [6]
Pyrexia	6.5% (5/77) [5]	8.3% (7/84) [7]	7.5% (12/161) [12]
Systemic inflammatory response syndrome	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Temperature intolerance	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Hepatobiliary Disorders	**2.6% (2/77) [2]**	**0.0% (0/84) [0]**	**1.2% (2/161) [2]**
Cholecystitis acute	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Cholestasis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Immune System Disorders	**2.6% (2/77) [2]**	**2.4% (2/84) [2]**	**2.5% (4/161) [4]**
Drug hypersensitivity	1.3% (1/77) [1]	2.4% (2/84) [2]	1.9% (3/161) [3]
Immune system disorder	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Infections and Infestations	**45.5% (35/77) [53]**	**38.1% (32/84) [42]**	**41.6% (67/161) [95]**
Acute sinusitis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Bacterial disease carrier	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Bronchitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Candida infection	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Cellulitis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Clostridium colitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Conjunctivitis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Cystitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Endocarditis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Fungal infection	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Fungal skin infection	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Gastroenteritis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Genital infection fungal	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Hand-foot-and-mouth disease	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Herpes zoster	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Oral candidiasis	3.9% (3/77) [3]	2.4% (2/84) [2]	3.1% (5/161) [5]
Parotitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Pneumonia	7.8% (6/77) [6]	9.5% (8/84) [8]	8.7% (14/161) [14]
Respiratory syncytial virus infection	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Respiratory tract infection	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Rhinitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Sepsis	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Septic shock	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Sinusitis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Skin candida	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Urinary tract infection	31.2% (24/77) [28]	20.2% (17/84) [19]	25.5% (41/161) [47]
Wound infection	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Injury, Poisoning and Procedural Complications	**10.4% (8/77) [8]**	**10.7% (9/84) [11]**	**10.6% (17/161) [19]**
Ankle fracture	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Carotid artery restenosis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]

Fall	3.9% (3/77) [3]	3.6% (3/84) [4]	3.7% (6/161) [7]
Joint dislocation	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Post procedural haematoma	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Procedural pain	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Procedural vomiting	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Pseudomeningocele	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Subdural haematoma	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Tracheal injury	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Vascular pseudoaneurysm	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Investigations	**14.3% (11/77) [12]**	**17.9% (15/84) [19]**	**16.1% (26/161) [31]**
Biopsy salivary gland	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Blood creatine phosphokinase increased	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Blood creatinine increased	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Blood glucose increased	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Blood magnesium abnormal	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Blood pressure increased	7.8% (6/77) [6]	9.5% (8/84) [9]	8.7% (14/161) [15]
Blood urea increased	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Electrocardiogram ST segment depression	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Glycosylated haemoglobin increased	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Transaminases increased	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Troponin increased	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Urine output decreased	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Metabolism and Nutrition Disorders	**40.3% (31/77) [40]**	**33.3% (28/84) [42]**	**36.6% (59/161) [82]**
Decreased appetite	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Dehydration	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]
Diabetes mellitus	3.9% (3/77) [3]	0.0% (0/84) [0]	1.9% (3/161) [3]
Dyslipidaemia	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Electrolyte imbalance	7.8% (6/77) [6]	4.8% (4/84) [4]	6.2% (10/161) [10]
Fluid retention	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Glucose tolerance impaired	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Gout	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Hypercholesterolaemia	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Hyperglycaemia	5.2% (4/77) [4]	4.8% (4/84) [4]	5.0% (8/161) [8]
Hyperlipidaemia	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Hypernatraemia	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Hypervolaemia	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Hypocalcaemia	2.6% (2/77) [2]	3.6% (3/84) [3]	3.1% (5/161) [5]
Hypokalaemia	5.2% (4/77) [4]	15.5% (13/84) [13]	10.6% (17/161) [17]
Hypomagnesaemia	5.2% (4/77) [4]	6.0% (5/84) [5]	5.6% (9/161) [9]
Hyponatraemia	2.6% (2/77) [2]	3.6% (3/84) [3]	3.1% (5/161) [5]
Hypophosphataemia	3.9% (3/77) [3]	1.2% (1/84) [1]	2.5% (4/161) [4]
Hypoproteinaemia	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Vitamin D deficiency	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Musculoskeletal and Connective Tissue Disorders	**20.8% (16/77) [20]**	**15.5% (13/84) [15]**	**18.0% (29/161) [35]**
Arthralgia	6.5% (5/77) [6]	3.6% (3/84) [4]	5.0% (8/161) [10]
Back pain	3.9% (3/77) [3]	3.6% (3/84) [3]	3.7% (6/161) [6]
Groin pain	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Joint swelling	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Muscle spasms	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]

Musculoskeletal chest pain	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Musculoskeletal pain	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]
Neck pain	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Osteoarthritis	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Pain in extremity	5.2% (4/77) [4]	3.6% (3/84) [3]	4.3% (7/161) [7]
Neoplasms Benign, Malignant and Unspecified (incl cysts and polyps)	**1.3% (1/77) [2]**	**0.0% (0/84) [0]**	**0.6% (1/161) [2]**
Appendix cancer	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Haemangioma of liver	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Nervous System Disorders	**66.2% (51/77) [90]**	**51.2% (43/84) [83]**	**58.4% (94/161) [173]**
Brain oedema	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Carotid artery dissection	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Carotid artery thrombosis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Cerebral artery embolism	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Cerebral haemorrhage	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Cerebral vasoconstriction	0.0% (0/77) [0]	13.1% (11/84) [11]	6.8% (11/161) [11]
Complex regional pain syndrome	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Convulsion	6.5% (5/77) [5]	1.2% (1/84) [1]	3.7% (6/161) [6]
Dizziness	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Encephalopathy	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Haemorrhagic transformation stroke	39.0% (30/77) [31]	29.8% (25/84) [26]	34.2% (55/161) [57]
Headache	15.6% (12/77) [12]	16.7% (14/84) [15]	16.1% (26/161) [27]
Hemianopia	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Horner's syndrome	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Hydrocephalus	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Hypertonia	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Intracranial pressure increased	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Intraventricular haemorrhage	3.9% (3/77) [3]	4.8% (4/84) [4]	4.3% (7/161) [7]
Ischaemic stroke	9.1% (7/77) [7]	3.6% (3/84) [4]	6.2% (10/161) [11]
Lethargy	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Migraine	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Monoparesis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Muscle spasticity	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Neuralgia	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Paraesthesia	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Partial seizures	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Somnolence	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Status epilepticus	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Stroke in evolution	10.4% (8/77) [8]	6.0% (5/84) [5]	8.1% (13/161) [13]
Subarachnoid haemorrhage	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Syncope	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Transient ischaemic attack	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Tremor	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Psychiatric Disorders	**32.5% (25/77) [29]**	**33.3% (28/84) [35]**	**32.9% (53/161) [64]**
Acute psychosis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Agitation	10.4% (8/77) [8]	7.1% (6/84) [6]	8.7% (14/161) [14]
Alcohol withdrawal syndrome	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Anxiety	3.9% (3/77) [3]	2.4% (2/84) [3]	3.1% (5/161) [6]
Confusional state	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Depressed mood	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Depression	14.3% (11/77) [11]	15.5% (13/84) [13]	14.9% (24/161) [24]
Depressive symptom	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Insomnia	5.2% (4/77) [4]	8.3% (7/84) [7]	6.8% (11/161) [11]
Mental status changes	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Restlessness	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Sleep disorder	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Stress	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Renal and Urinary Disorders	**19.5% (15/77) [16]**	**16.7% (14/84) [16]**	**18.0% (29/161) [32]**
Dysuria	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Haematuria	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Hypertonic bladder	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Oliguria	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Polyuria	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Renal failure	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Renal failure acute	3.9% (3/77) [3]	4.8% (4/84) [4]	4.3% (7/161) [7]
Renal infarct	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Renal mass	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Urethral pain	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Urge incontinence	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Urinary retention	3.9% (3/77) [3]	8.3% (7/84) [7]	6.2% (10/161) [10]
Reproductive System and Breast Disorders	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Benign prostatic hyperplasia	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Respiratory, Thoracic and Mediastinal Disorders	29.9% (23/77) [27]	26.2% (22/84) [32]	28.0% (45/161) [59]
Acute respiratory failure	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Aspiration	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Atelectasis	2.6% (2/77) [2]	3.6% (3/84) [3]	3.1% (5/161) [5]
Chronic obstructive pulmonary disease	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Cough	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Dyspnoea	1.3% (1/77) [1]	2.4% (2/84) [2]	1.9% (3/161) [3]
Haemoptysis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Hiccups	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Hypoxia	1.3% (1/77) [1]	2.4% (2/84) [2]	1.9% (3/161) [3]
Oropharyngeal pain	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Pleural effusion	1.3% (1/77) [1]	4.8% (4/84) [4]	3.1% (5/161) [5]
Pneumonia aspiration	6.5% (5/77) [5]	7.1% (6/84) [6]	6.8% (11/161) [11]
Pulmonary congestion	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Pulmonary embolism	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Pulmonary oedema	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Respiratory arrest	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Respiratory distress	5.2% (4/77) [4]	1.2% (1/84) [1]	3.1% (5/161) [5]
Respiratory failure	2.6% (2/77) [2]	3.6% (3/84) [3]	3.1% (5/161) [5]
Sleep apnoea syndrome	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Wheezing	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Skin and Subcutaneous Tissue Disorders	**13.0% (10/77) [10]**	**11.9% (10/84) [10]**	**12.4% (20/161) [20]**
Decubitus ulcer	0.0% (0/77) [0]	4.8% (4/84) [4]	2.5% (4/161) [4]
Dermatitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Dermatitis contact	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Ecchymosis	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]

Erythema	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]
Intertrigo	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Pruritus	1.3% (1/77) [1]	1.2% (1/84) [1]	1.2% (2/161) [2]
Rash	1.3% (1/77) [1]	2.4% (2/84) [2]	1.9% (3/161) [3]
Rash erythematous	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Skin lesion	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Surgical and Medical Procedures	**1.3% (1/77) [1]**	**0.0% (0/84) [0]**	**0.6% (1/161) [1]**
Gastrostomy tube removal	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Vascular Disorders	**20.8% (16/77) [18]**	**25.0% (21/84) [26]**	**23.0% (37/161) [44]**
Aortic aneurysm	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Arterial rupture	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Deep vein thrombosis	7.8% (6/77) [6]	6.0% (5/84) [5]	6.8% (11/161) [11]
Femoral artery embolism	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Haematoma	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Hypertension	3.9% (3/77) [3]	6.0% (5/84) [5]	5.0% (8/161) [8]
Hypotension	5.2% (4/77) [4]	6.0% (5/84) [5]	5.6% (9/161) [9]
Lymphoedema	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Phlebitis	1.3% (1/77) [1]	0.0% (0/84) [0]	0.6% (1/161) [1]
Thrombophlebitis	1.3% (1/77) [1]	3.6% (3/84) [3]	2.5% (4/161) [4]
Thrombosis	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Vascular occlusion	0.0% (0/77) [0]	1.2% (1/84) [1]	0.6% (1/161) [1]
Vasospasm	0.0% (0/77) [0]	3.6% (3/84) [3]	1.9% (3/161) [3]

Table 10. CEC Adjudicated Serious Adverse Events (Analysis Cohort)

MedDRA Preferred Term	IV t-PA Only	IV t-PA + Solitaire™	All Enrolled
Units	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]
TOTAL Adverse Events (AE)	**33.8% (26/77) [54]**	**31.0% (26/84) [44]**	**32.3% (52/161) [98]**
Blood and Lymphatic System Disorders	**0.0% (0/77) [0]**	**2.4% (2/84) [2]**	**1.2% (2/161) [2]**
Anaemia	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Cardiac Disorders	**5.2% (4/77) [5]**	**8.3% (7/84) [8]**	**6.8% (11/161) [13]**
Cardiac Arrest	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Intracardiac thrombus	0.0% (0/77) [0]	2.4% (2/84) [2]	1.2% (2/161) [2]
Tachycardia	2.6% (2/77) [2]	0.0% (0/84) [0]	1.2% (2/161) [2]
Gastrointestinal Disorders	**2.6% (2/77) [2]**	**2.4% (2/84) [2]**	**2.5% (4/161) [4]**
Infections and Infestations	**7.8% (6/77) [7]**	**3.6% (3/84) [3]**	**5.6% (9/161) [10]**
Sepsis	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Injury, Poisoning and Procedural Complications	**3.9% (3/77) [3]**	**3.6% (3/84) [3]**	**3.7% (6/161) [6]**
Nervous System Disorders	**20.8% (16/77) [21]**	**11.9% (10/84) [12]**	**16.1% (26/161) [33]**
Brain oedema	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Haemorrhagic transformation stroke	5.2% (4/77) [4]	0.0% (0/84) [0]	2.5% (4/161) [4]
Ischaemic stroke	6.5% (5/77) [5]	2.4% (2/84) [3]	4.3% (7/161) [8]
Stroke in evolution	10.4% (8/77) [8]	4.8% (4/84) [4]	7.5% (12/161) [12]
Renal and Urinary Disorders	**3.9% (3/77) [3]**	**3.6% (3/84) [3]**	**3.7% (6/161) [6]**
Renal failure acute	1.3% (1/77) [1]	2.4% (2/84) [2]	1.9% (3/161) [3]
Respiratory, Thoracic and Mediastinal Disorders	**9.1% (7/77) [8]**	**7.1% (6/84) [7]**	**8.1% (13/161) [15]**
Respiratory distress	3.9% (3/77) [3]	0.0% (0/84) [0]	1.9% (3/161) [3]
Respiratory failure	2.6% (2/77) [2]	2.4% (2/84) [2]	2.5% (4/161) [4]
Vascular Disorders	**5.2% (4/77) [4]**	**3.6% (3/84) [3]**	**4.3% (7/161) [7]**
Deep vein thrombosis	2.6% (2/77) [2]	1.2% (1/84) [1]	1.9% (3/161) [3]

Table 11. CEC-Adjudicated Device-Related Adverse Events

Terminology	Severity	Outcome	Neurological deterioration or death
Subarachnoid Contrast Extravasation	Mild	Recovered without sequelae	None
Cerebral Vasospasm	Moderate	Recovered without sequelae	None
Intraventricular Hemorrhage	Mild	Recovered without sequelae	None
Subarachnoid Hemorrhage	Mild	Recovered without sequelae	None
Cerebral Vasospasm	Mild	Recovered without sequelae	None
Cerebral Vasospasm	Moderate	Recovered without sequelae	Brain edema 1 day post-ADE, alive through study follow-up

Table 12. Screen Failure	
Reason for Screen Failure	Subjects (no.)
Thrombolysis in Cerebral Infarction (TICI) > 1 flow in the intracranial internal carotid artery, M1 segment of the MCA, or carotid terminus confirmed by CT or MR angiography that is accessible to the Solitaire™ FR Device.	28
MRI- or CT-assessed core infarct lesion greater than 50 cc; severe hypoperfusion lesion (Tmax>10secs lesion greater than 100 cc); and/or Ischemic penumbra < 15 cc and mismatch ratio ≤1.8.	17
CT showing hypodensity or MRI showing hyperintensity involving greater than 1/3 of the middle cerebral artery (MCA) territory (or in other territories, >100 cc of tissue) on presentation.	6
NIHSS < 8 or ≥ 30 at the time of randomization	9
CTA or MRA evidence of carotid dissection or complete cervical carotid occlusion.	3
CT or MRI evidence of mass effect or intra-cranial tumor (except small meningioma).	3
Rapid neurological improvement prior to study randomization suggesting resolution of signs/ symptoms of stroke.	4
Pre-stroke Modified Rankin Score> 1	2
Baseline non-contrast CT or DWI MRI evidence of a moderate/large core defined as extensive early ischemic changes of Alberta Stroke Program Early CT score (ASPECTS) < 6.	3
Imaging evidence that suggests, in the opinion of the investigator, the subject is not appropriate for mechanical thrombectomy intervention (e.g., inability to navigate to target lesion, moderate/large infarct with poor collateral circulation, etc.).	3
Previous intracranial hemorrhage, neoplasm, subarachnoid hemorrhage, cerebral aneurysm, or arteriovenous malformation.	2
CTA or MRA evidence of carotid dissection or complete cervical carotid occlusion requiring stenting at the time of the index procedure (i.e., mechanical thrombectomy).	3
Subject is unwilling to conduct protocol-required follow-up visits.	1
Age >80 years old	1
Known hereditary or acquired hemorrhagic diathesis, coagulation factor deficiency.	1
Contraindication to IV t-PA as per local national guidelines.	1
History of stroke in the past 3 months.	1
Arterial tortuosity, calcification, pre-existing stent, and/or stenosis which would prevent the device from reaching the target vessel and/or preclude safe recovery of the device.	2
Subject is unable to be treated within 6 hours of onset of stroke symptoms and within 1.5 hours (90 minutes) from qualifying imaging to groin puncture.	2
CT or MRI evidence of a basilar artery (BA) occlusion or posterior cerebral artery (PCA) occlusion.	1
No certified rater available to assess ASPECTS prior to study enrollment	1
Total	**77***
*** Column does not sum to 77 due to some subjects having more than 1 screen failure reason**	

SAFETY AND EFFECTIVENESS INFORMATION – INDICATION 2

SWIFT Clinical Study

SWIFT (Solitaire™ With the intention For Thrombectomy study; IDE G090082) is a multicenter, randomized, prospectively controlled study in clot retrieval for patients diagnosed with an acute ischemic stroke. The objective of the study is to demonstrate substantial equivalence of the Solitaire™ FR Revascularization Device with the Concentric Medical Merci™ device. Primary efficacy was demonstrated by arterial recanalization of occluded target vessel measured by Thrombolysis in Myocardial Infarction (TIMI) score of 2 or 3 following the use of the Solitaire™ device without any symptomatic intracranial hemorrhage.

Inclusion Criteria

Subjects were considered eligible for the study if they satisfied all of the following inclusion criteria:

- Subject or subject's legally authorized representative has signed and dated an Informed Consent Form
- Age 22 – 85
- Clinical signs consistent with acute ischemic stroke
- National Institute of Health Stoke Scale (NIHSS) score of at least 8 and less than 30
- Thrombolysis in Myocardial Infarction (TIMI) 0 or TIMI 1 flow in the M1 or M2 of MCA, ICA, basilar or vertebral arteries confirmed by angiography that is accessible to the Solitaire™ FR device or Concentric Medical's MERCI device
- Subject is able to be treated within 8 hours of stroke symptoms onset with minimum of one deployment of the Solitaire™ FR device or Concentric Medical's MERCI device
- Subject who is ineligible or failed intravenous tissue plasminogen activator (t-PA) therapy given at local institutions or within national practice. (**NOTE**: Failed IV-tPA is defined as subjects with occlusion present 61 minutes or more after start of IV therapy)
- Subject is willing to conduct follow-up visits

Exclusion Criteria

Subjects were considered ineligible for study participation if they met any of the following exclusion criteria:

- NIHSS score of 30 or greater
- Neurological signs that are rapidly improving prior to or at time of treatment
- Females who are pregnant or lactating
- Known serious sensitivity to radiographic contrast agents
- Current participation in another investigational drug or device study
- Uncontrolled hypertension defined as systolic blood pressure >185 or diastolic blood pressure >110 that cannot be controlled except with continuous parenteral antihypertensive medication
- Use of warfarin anticoagulation with INR > 3.0
- Platelet count < 30,000
- Glucose < 50 mg/dl
- Arterial tortuosity that would prevent the device from reaching the target vessel
- Life expectancy of less than 90 days
- Additionally, subjects were also considered ineligible for study participation if they met any of the following imaging exclusion criteria:
  - CT or MRI evidence of hemorrhage on presentation
  - CT showing hypodensity or MR showing hyperintensity involving greater than 1/3 of the middle cerebral artery (MCA) territory (or in other territories, >100 cc of tissue) on presentation
  - CT or MRI evidence of mass effect or intracranial tumor (except small meningioma)
  - Angiographic evidence of carotid dissection, complete cervical carotid occlusions, or vasculitis

Study Endpoints

The primary effectiveness endpoint of the study is successful recanalization with no symptomatic hemorrhage (SRNH), defined as achieving TIMI 2 or 3 flow in all treatable vessels without any presence of symptomatic intracranial hemorrhage.

- Successful recanalization is defined as achieving TIMI 2 or 3 flow in all treatable vessels.
- Successful recanalization of the MCA includes all M1 and M2 segments, internal carotid terminus lesions include the ICA, M1 and both M2 branches, and posterior circulation includes both the vertebral and basilar arteries.

Symptomatic intracranial hemorrhage is defined as any PH1, PH2, RIH, SAH, or IVH associated with a decline in NIHSS ≥ 4 within 24hrs.

Results

Fifty eight (58) randomized Solitaire™ FR Revascularization Device and thirty one (31) roll-in patients were treated with the Solitaire™ FR Revascularization device in the SWIFT study. The median age and NIHSS at the time of treatment were 66 years and 17, respectively. Data from this study demonstrate that the Solitaire™ FR Revascularization Device is non-inferior to the Merci device for arterial recanalization of the occluded target vessel without any presence of symptomatic intracranial hemorrhage. A summary of the subject disposition is presented below.

Table 13. Summary of Subject Disposition

Subject Disposition	Roll-in (all Solitaire™ FR device)	Randomized Solitaire™ FR device	Randomized Concentric Medical MERCI™ device	Overall
Completed Study	22/31 (71%)	45/58 (78%)	27/55 (49%)	94/144 (65%)
Death	5/31 (16%)	10/58 (17%)	21/55 (38%)	36/144 (25%)
Early Terminated				
Subject Lost to Follow-Up	3/31 (10%)	2/58 (3%)	0/55 (0%)	5/144 (3%)
Subject Withdrawal	0/31 (0%)	0/58 (0%)	2/55 (4%)	2/144 (1%)
Other	1/31 (3%)	1/58 (1%)	5/55 (5%)	7/144 (3%)
Total	31 (100%)	58 (100%)	55/55 (100%)	144/144 (100%)

All subjects who were consented and randomized were included in the intention-to-treat (ITT) population. The ITT population included all subjects with data for a given endpoint; results were assessed according to randomized assignment regardless of the treatment actually received.

The primary efficacy analysis (successful recanalization measured by TIMI flow without symptomatic hemorrhage) was performed using a one-sided test under Blackwelder's method of testing non-inferiority at the 0.025 level of significance. The primary safety endpoint of device- and/or procedure related Serious Adverse Events (SAEs) was evaluated descriptively.

NOTE: The SFR4-3-20-10, SFR4-3-40-10, SFR4-4-20-05, SFR4-4-20-10, SFR4-4-40-10, SFR4-6-20-10, SFR4-6-24-06 and SFR4-6-40-10 models were not evaluated as part of the SWIFT study.

Table 14. Primary Effectiveness Endpoint– Successful Revascularization without Symptomatic Hemorrhage (ITT)

Success Criteria	Randomized Solitaire™ FR device % (n/N) [95% CI][1] N= 58	Randomized Concentric Medical Merci™ device % (n/N) [95% CI][1] N= 55	Randomized Difference [95% CI][2]	Randomized. Non-inferiority $\Delta = 0.10$ p-value[3]	Randomized Superiority p-value[4]
Successful revascularization (TIMI 2/3) without use of rescue therapy or incidence of protocol-defined bleed events	60.7% (34/56)[5] [45.7%, 74.4%]	24.1% (13/54)[5] [12.8%, 38.8%]	36.6% [16.9%, 54.7%]	<0.0001	0.0003

Table 14. Primary Effectiveness Endpoint– Successful Revascularization without Symptomatic Hemorrhage (ITT)					
Success Criteria	**Randomized Solitaire™ FR device % (n/N) [95% CI][1] N= 58**	**Randomized Concentric Medical Merci™ device % (n/N) [95% CI][1] N= 55**	**Randomized Difference [95% CI][2]**	**Randomized. Non-inferiority Δ = 0.10 p-value[3]**	**Randomized Superiority p-value[4]**

[1] Clopper-Pearson method for estimating exact binomial confidence intervals adjusted for interim analysis.

[2] Exact Unconditional Confidence Limits for the Risk Difference adjusted for interim analysis.

[3] One-sided non-inferiority test based on Wald method at α=0.025 adjusted for interim analysis.

[4] Two-sided Fisher's Exact p-value adjusted for interim analysis.

[5] 2 subjects from the randomized Solitaire™ FR cohort and 1 from the randomized Concentric Medical Merci™ device cohorts were not included in the analysis because imaging data was not available from assessment by the Core Lab.

Table 15. Major Secondary Effectiveness Endpoint – Time to Achieve Initial Recanalization (ITT)			
Characteristic	**Randomized Solitaire™ FR device Mean ± SD [Median] (min, max)**	**Concentric Medical Merci™ device Mean ± SD [Median] (min, max)**	**Randomized p-value[1]**
Time from mechanical thrombectomy start to first TIMI 2 or 3 flow (minutes) (failures of revascularization - total procedure time)[2]	47.0 ± 37.3 [36.0] (5.0, 178.0) (n=47)	58.7 ± 33.8 [52.0] (13.0, 160.0) (n=46)	**0.0376**

[1] P-values obtained from Wilcoxon's rank-sum test for nonparametric variables.

[2] Patient who failed to achieve recanalization had total procedure time included.

Unavailability of data contributed to fewer subjects analyzed compared to the cohort sample size.

Table 16. Primary Safety Endpoint – Incidence of Device- and Procedure-Related Serious Adverse Events (ITT)				
Characteristic	**Roll – In (all Solitaire™ FR) N=31**	**Rand Solitaire™ FR N=58**	**Randomized Concentric Medical MERCI™ device**	**Rand. p-value**
Study Device Related	12.9% (4/31) [5]	8.6% (5/58) [7]	16.4% (9/55) [13]	0.2601
Ancillary Device Related	9.7% (3/31) [3]	5.2% (3/58) [3]	3.6% (2/55) [2]	1.0000
Unknown Device Related[1]	3.2% (1/31) [1]	1.7% (1/58) [1]	10.9% (6/55) [6]	0.0566
Procedure/Treatment Related	19.4% (6/31) [9]	13.8% (8/58) [17]	16.4% (9/55) [12]	0.7950

[1] CEC adjudicated event as Device related but unable to attribute to study device or ancillary device. Ancillary device includes microcatheter, guidewire, guide catheter, or another device that is not the study device.

Table 17. CEC-Adjudicated Adverse Events ITT				
MedDRA Preferred Term	**Roll-in (all Solitaire™ FR device)**	**Randomized Solitaire™ FR device**	**Randomized Concentric Medical MERCI™ device**	**All Enrolled**
Units	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]
TOTAL Adverse Events (AE)	**83.9% (26/31) [115]**	**94.8% (55/58) [267]**	**92.7% (51/55) [262]**	**91.7% (132/144) [644]**
Blood and Lymphatic System Disorders	**22.6% (7/31) [8]**	**24.1% (14/58) [17]**	**16.4% (9/55) [9]**	**20.8% (30/144) [34]**
Anaemia	19.4% (6/31) [6]	20.7% (12/58) [13]	12.7% (7/55) [7]	17.4% (25/144) [26]
Thrombocytopenia	3.2% (1/31) [1]	6.9% (4/58) [4]	1.8% (1/55) [1]	4.2% (6/144) [6]
Cardiac Disorders	**22.6% (7/31) [7]**	**39.7% (23/58) [28]**	**32.7% (18/55) [22]**	**33.3% (48/144) [57]**
Atrial Fibrillation	16.1% (5/31) [5]	19.0% (11/58) [12]	9.1% (5/55) [5]	14.6% (21/144) [22]
Bradycardia	0.0% (0/31) [0]	1.7% (1/58) [1]	5.5% (3/55) [3]	2.8% (4/144) [4]
Cardiac Arrest	0.0% (0/31) [0]	0.0% (0/58) [0]	5.5% (3/55) [4]	2.1% (3/144) [4]
Cardiac Failure Congestive	3.2% (1/31) [1]	5.2% (3/58) [3]	3.6% (2/55) [2]	4.2% (6/144) [6]
Gastrointestinal Disorders	**16.1% (5/31) [7]**	**20.7% (12/58) [16]**	**21.8% (12/55) [18]**	**20.1% (29/144) [41]**
Dysphagia	0.0% (0/31) [0]	5.2% (3/58) [3]	7.3% (4/55) [4]	4.9% (7/144) [7]
Gastrointestinal Haemorrhage	9.7% (3/31) [3]	1.7% (1/58) [1]	1.8% (1/55) [2]	3.5% (5/144) [6]
Nausea	6.5% (2/31) [2]	1.7% (1/58) [1]	10.9% (6/55) [8]	6.3% (9/144) [11]
General Disorders and Administration Site Conditions	**9.7% (3/31) [3]**	**24.1% (14/58) [15]**	**16.4% (9/55) [10]**	**18.1% (26/144) [28]**
Catheter Site Haematoma	9.7% (3/31) [3]	5.2% (3/58) [3]	1.8% (1/55) [1]	4.9% (7/144) [7]
Pyrexia	0.0% (0/31) [0]	12.1% (7/58) [7]	10.9% (6/55) [6]	9.0% (13/144) [13]
Infections and Infestations	**41.9% (13/31) [16]**	**43.1% (25/58) [29]**	**34.5% (19/55) [32]**	**39.6% (57/144) [77]**
Pneumonia	9.7% (3/31) [3]	3.4% (2/58) [2]	12.7% (7/55) [7]	8.3% (12/144) [12]
Sepsis	0.0% (0/31) [0]	5.2% (3/58) [3]	3.6% (2/55) [2]	3.5% (5/144) [5]
Urinary Tract Infection	32.3% (10/31) [10]	25.9% (15/58) [15]	21.8% (12/55) [13]	25.7% (37/144) [38]
Metabolism and Nutrition Disorders	**12.9% (4/31) [5]**	**20.7% (12/58) [16]**	**21.8% (12/55) [24]**	**19.4% (28/144) [45]**
Electrolyte Imbalance	0.0% (0/31) [0]	5.2% (3/58) [3]	3.6% (2/55) [2]	3.5% (5/144) [5]
Hyperglycaemia	6.5% (2/31) [2]	3.4% (2/58) [2]	7.3% (4/55) [5]	5.6% (8/144) [9]
Hypernatraemia	0.0% (0/31) [0]	3.4% (2/58) [2]	5.5% (3/55) [3]	3.5% (5/144) [5]
Hypocalcaemia	3.2% (1/31) [1]	3.4% (2/58) [2]	5.5% (3/55) [3]	4.2% (6/144) [6]
Hypokalaemia	3.2% (1/31) [1]	0.0% (0/58) [0]	7.3% (4/55) [5]	3.5% (5/144) [6]
Musculoskeletal and Connective Tissue Disorders	**12.9% (4/31) [5]**	**13.8% (8/58) [8]**	**7.3% (4/55) [4]**	**11.1% (16/144) [17]**
Musculoskeletal Pain	6.5% (2/31) [3]	1.7% (1/58) [1]	3.6% (2/55) [2]	3.5% (5/144) [6]
Nervous System Disorders	**58.1% (18/31) [30]**	**55.2% (32/58) [51]**	**74.5% (41/55) [70]**	**63.2% (91/144) [151]**
Brain Oedema	6.5% (2/31) [2]	5.2% (3/58) [3]	1.8% (1/55) [1]	4.2% (6/144) [6]
Cerebral Gas Embolism	0.0% (0/31) [0]	1.7% (1/58) [1]	5.5% (3/55) [3]	2.8% (4/144) [4]
Cerebral Haematoma	12.9% (4/31) [4]	10.3% (6/58) [6]	18.2% (10/55) [10]	13.9% (20/144) [20]
Cerebrovascular Accident	19.4% (6/31) [6]	12.1% (7/58) [7]	25.5% (14/55) [14]	18.8% (27/144) [27]
Cerebrovascular Spasm	6.5% (2/31) [2]	5.2% (3/58) [3]	7.3% (4/55) [4]	6.3% (9/144) [9]

Haemorrhagic Cerebral Infarction	25.8% (8/31) [8]	12.1% (7/58) [7]	9.1% (5/55) [5]	13.9% (20/144) [20]
Headache	9.7% (3/31) [3]	10.3% (6/58) [6]	3.6% (2/55) [2]	7.6% (11/144) [11]
Intracranial Pressure Increased	0.0% (0/31) [0]	6.9% (4/58) [4]	5.5% (3/55) [3]	4.9% (7/144) [7]
Intraventricular Haemorrhage	0.0% (0/31) [0]	3.4% (2/58) [2]	5.5% (3/55) [3]	3.5% (5/144) [5]
Ischaemic Stroke	6.5% (2/31) [2]	3.4% (2/58) [2]	12.7% (7/55) [7]	7.6% (11/144) [11]
Subarachnoid Haemorrhage	3.2% (1/31) [1]	5.2% (3/58) [3]	14.5% (8/55) [8]	8.3% (12/144) [12]
Respiratory, Thoracic and Mediastinal Disorders	**25.8% (8/31) [12]**	**37.9% (22/58) [32]**	**41.8% (23/55) [29]**	**36.8% (53/144) [73]**
Atelectasis	0.0% (0/31) [0]	6.9% (4/58) [4]	3.6% (2/55) [2]	4.2% (6/144) [6]
Hypoxia	0.0% (0/31) [0]	10.3% (6/58) [6]	1.8% (1/55) [1]	4.9% (7/144) [7]
Pleural Effusion	9.7% (3/31) [3]	0.0% (0/58) [0]	7.3% (4/55) [4]	4.9% (7/144) [7]
Pneumonia Aspiration	3.2% (1/31) [1]	10.3% (6/58) [6]	5.5% (3/55) [3]	6.9% (10/144) [10]
Pulmonary Oedema	0.0% (0/31) [0]	5.2% (3/58) [3]	10.9% (6/55) [6]	6.3% (9/144) [9]
Respiratory Failure	9.7% (3/31) [3]	8.6% (5/58) [5]	14.5% (8/55) [8]	11.1% (16/144) [16]
Vascular Disorders	**32.3% (10/31) [11]**	**27.6% (16/58) [24]**	**32.7% (18/55) [21]**	**30.6% (44/144) [56]**
Deep Vein Thrombosis	12.9% (4/31) [4]	5.2% (3/58) [3]	9.1% (5/55) [5]	8.3% (12/144) [12]
Hypertension	0.0% (0/31) [0]	10.3% (6/58) [7]	1.8% (1/55) [1]	4.9% (7/144) [8]
Hypotension	9.7% (3/31) [3]	10.3% (6/58) [7]	10.9% (6/55) [6]	10.4% (15/144) [16]
Vasospasm	9.7% (3/31) [3]	5.2% (3/58) [3]	5.5% (3/55) [3]	6.3% (9/144) [9]

Table 18. CEC-Adjudicated Serious Adverse Events (ITT)

MedDRA Preferred Term	Roll-in (all Solitaire™ FR device) N=31	Randomized Solitaire™ FR device N=58	Randomized Concentric Medical MERCI™ device N = 55	All Enrolled N=144
Units	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]
TOTAL Serious Adverse Events (SAE)	**77.4% (24/31) [67]**	**84.5% (49/58) [155]**	**80.0% (44/55) [169]**	**81.3% (117/144) [391]**
Blood and Lymphatic System Disorders	**9.7% (3/31) [3]**	**15.5% (9/58) [11]**	**10.9% (6/55) [6]**	**12.5% (18/144) [20]**
Anaemia	9.7% (3/31) [3]	15.5% (9/58) [10]	7.3% (4/55) [4]	11.1% (16/144) [17]
Cardiac Disorders	**19.4% (6/31) [6]**	**31.0% (18/58) [22]**	**25.5% (14/55) [18]**	**26.4% (38/144) [46]**
Atrial fibrillation	16.1% (5/31) [5]	19.0% (11/58) [12]	7.3% (4/55) [4]	13.9% (20/144) [21]
Bradycardia	0.0% (0/31) [0]	1.7% (1/58) [1]	3.6% (2/55) [2]	2.1% (3/144) [3]
Cardiac arrest	0.0% (0/31) [0]	0.0% (0/58) [0]	5.5% (3/55) [4]	2.1% (3/144) [4]
Cardiac failure congestive	3.2% (1/31) [1]	3.4% (2/58) [2]	1.8% (1/55) [1]	2.8% (4/144) [4]
Sinus bradycardia	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]	1.4% (2/144) [2]
Eye Disorders	**3.2% (1/31) [1]**	**1.7% (1/58) [1]**	**1.8% (1/55) [1]**	**2.1% (3/144) [3]**
Retinal artery embolism	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Gastrointestinal Disorders	**16.1% (5/31) [6]**	**12.1% (7/58) [9]**	**14.5% (8/55) [10]**	**13.9% (20/144) [25]**
Abdominal pain	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Dysphagia	0.0% (0/31) [0]	1.7% (1/58) [1]	7.3% (4/55) [4]	3.5% (5/144) [5]
Gastrointestinal haemorrhage	9.7% (3/31) [3]	1.7% (1/58) [1]	1.8% (1/55) [2]	3.5% (5/144) [6]
Haematochezia	0.0% (0/31) [0]	3.4% (2/58) [2]	0.0% (0/55) [0]	1.4% (2/144) [2]
Nausea	3.2% (1/31) [1]	0.0% (0/58) [0]	5.5% (3/55) [3]	2.8% (4/144) [4]
Upper gastrointestinal haemorrhage	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
General Disorders and Administration Site Conditions	**3.2% (1/31) [1]**	**10.3% (6/58) [7]**	**7.3% (4/55) [4]**	**7.6% (11/144) [12]**
Catheter site haematoma	3.2% (1/31) [1]	3.4% (2/58) [2]	0.0% (0/55) [0]	2.1% (3/144) [3]
Pyrexia	0.0% (0/31) [0]	5.2% (3/58) [3]	5.5% (3/55) [3]	4.2% (6/144) [6]
Infections and Infestations	**12.9% (4/31) [6]**	**19.0% (11/58) [12]**	**27.3% (15/55) [20]**	**20.8% (30/144) [38]**
Bacteraemia	3.2% (1/31) [1]	1.7% (1/58) [1]	3.6% (2/55) [2]	2.8% (4/144) [4]
Clostridial infection	3.2% (1/31) [1]	1.7% (1/58) [1]	1.8% (1/55) [1]	2.1% (3/144) [3]
Pneumonia	9.7% (3/31) [3]	3.4% (2/58) [2]	9.1% (5/55) [5]	6.9% (10/144) [10]
Sepsis	0.0% (0/31) [0]	5.2% (3/58) [3]	3.6% (2/55) [2]	3.5% (5/144) [5]
Urinary tract infection	3.2% (1/31) [1]	5.2% (3/58) [3]	10.9% (6/55) [6]	6.9% (10/144) [10]
Injury, Poisoning and Procedural Complications	**6.5% (2/31) [2]**	**6.9% (4/58) [4]**	**0.0% (0/55) [0]**	**4.2% (6/144) [6]**
Pneumothorax traumatic	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Vascular access complication	3.2% (1/31) [1]	1.7% (1/58) [1]	0.0% (0/55) [0]	1.4% (2/144) [2]
Investigations	**3.2% (1/31) [1]**	**10.3% (6/58) [6]**	**3.6% (2/55) [2]**	**6.3% (9/144) [9]**
Echocardiogram abnormal	0.0% (0/31) [0]	3.4% (2/58) [2]	0.0% (0/55) [0]	1.4% (2/144) [2]
International normalized ratio increased	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Metabolism and Nutrition Disorders	**3.2% (1/31) [2]**	**3.4% (2/58) [2]**	**5.5% (3/55) [3]**	**4.2% (6/144) [7]**
Diabetes mellitus	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Hyperglycaemia	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Musculoskeletal and Connective Tissue Disorders	**6.5% (2/31) [2]**	**1.7% (1/58) [1]**	**1.8% (1/55) [1]**	**2.8% (4/144) [4]**
Compartment syndrome	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Musculoskeletal pain	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Nervous System Disorders	**45.2% (14/31) [18]**	**43.1% (25/58) [34]**	**60.0% (33/55) [59]**	**50.0% (72/144) [111]**
Brain oedema	6.5% (2/31) [2]	3.4% (2/58) [2]	1.8% (1/55) [1]	3.5% (5/144) [5]
Carotid artery dissection	3.2% (1/31) [1]	0.0% (0/58) [0]	1.8% (1/55) [1]	1.4% (2/144) [2]
Cerebral gas embolism	0.0% (0/31) [0]	1.7% (1/58) [1]	5.5% (3/55) [3]	2.8% (4/144) [4]
Cerebral haematoma	6.5% (2/31) [2]	8.6% (5/58) [5]	14.5% (8/55) [8]	10.4% (15/144) [15]
Cerebrovascular accident	19.4% (6/31) [6]	12.1% (7/58) [7]	23.6% (13/55) [13]	18.1% (26/144) [26]
Cerebrovascular spasm	6.5% (2/31) [2]	5.2% (3/58) [3]	7.3% (4/55) [4]	6.3% (9/144) [9]
Clonus	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Convulsion	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]	1.4% (2/144) [2]
Depressed level of consciousness	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]	1.4% (2/144) [2]
Haemorrhagic cerebral infarction	6.5% (2/31) [2]	5.2% (3/58) [3]	5.5% (3/55) [3]	5.6% (8/144) [8]
Intracranial pressure increased	0.0% (0/31) [0]	3.4% (2/58) [2]	3.6% (2/55) [2]	2.8% (4/144) [4]
Intraventricular haemorrhage	0.0% (0/31) [0]	3.4% (2/58) [2]	3.6% (2/55) [2]	2.8% (4/144) [4]
Ischaemic stroke	3.2% (1/31) [1]	3.4% (2/58) [2]	12.7% (7/55) [7]	6.9% (10/144) [10]
Subarachnoid haemorrhage	3.2% (1/31) [1]	5.2% (3/58) [3]	12.7% (7/55) [7]	7.6% (11/144) [11]
Psychiatric Disorders	**0.0% (0/31) [0]**	**3.4% (2/58) [3]**	**0.0% (0/55) [0]**	**1.4% (2/144) [3]**
Agitation	0.0% (0/31) [0]	3.4% (2/58) [2]	0.0% (0/55) [0]	1.4% (2/144) [2]
Renal and Urinary Disorders	**0.0% (0/31) [0]**	**3.4% (2/58) [2]**	**5.5% (3/55) [3]**	**3.5% (5/144) [5]**
Respiratory, Thoracic and Mediastinal Disorders	**19.4% (6/31) [9]**	**25.9% (15/58) [20]**	**34.5% (19/55) [25]**	**27.8% (40/144) [54]**
Dyspnoea	3.2% (1/31) [1]	1.7% (1/58) [1]	3.6% (2/55) [2]	2.8% (4/144) [4]
Hypoxia	0.0% (0/31) [0]	6.9% (4/58) [4]	1.8% (1/55) [1]	3.5% (5/144) [5]
Pleural effusion	6.5% (2/31) [2]	0.0% (0/58) [0]	5.5% (3/55) [3]	3.5% (5/144) [5]

Pneumonia aspiration	3.2% (1/31) [1]	10.3% (6/58) [6]	5.5% (3/55) [3]	6.9% (10/144) [10]
Pneumothorax	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Pulmonary oedema	0.0% (0/31) [0]	3.4% (2/58) [2]	9.1% (5/55) [5]	4.9% (7/144) [7]
Respiratory distress	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]	0.7% (1/144) [1]
Respiratory failure	9.7% (3/31) [3]	8.6% (5/58) [5]	14.5% (8/55) [8]	11.1% (16/144) [16]
Vascular Disorders	**32.3% (10/31) [10]**	**24.1% (14/58) [20]**	**23.6% (13/55) [15]**	**25.7% (37/144) [45]**
Deep vein thrombosis	12.9% (4/31) [4]	5.2% (3/58) [3]	3.6% (2/55) [2]	6.3% (9/144) [9]
Hypertension	0.0% (0/31) [0]	6.9% (4/58) [4]	1.8% (1/55) [1]	3.5% (5/144) [5]
Hypotension	9.7% (3/31) [3]	10.3% (6/58) [7]	9.1% (5/55) [5]	9.7% (14/144) [15]
Vasospasm	9.7% (3/31) [3]	5.2% (3/58) [3]	5.5% (3/55) [3]	6.3% (9/144) [9]
Vessel perforation	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]	1.4% (2/144) [2]

NOTE: Table includes Serious Adverse Events occurring at an incidence of 2% or more in any group in the SWIFT Trial.

Table 19. Mortality at 90 Days (ITT)

Outcome	Rand. Solitaire™ FR % (n/N) [95% CI][1] N=58	Rand. Concentric Medical MERCI™ device % (n/N) [95% CI][1] N=55	Rand. Difference [95% CI][2]	Test for Non-inferiority $\Delta = 0.10$ p-value[3]	Test for Superiority p-value[4]
Mortality (ITT)	17.2% (10/58) [8.6%, 29.4%]	38.2% (31/55) [25.4%, 52.3%]	20.9% [2.7%, 38.6%]	0.0001	0.0196
Non-fatal-stroke-related morbidity	43.1% (25/28) [30.2%, 56.8%]	32.7% (18/55) [20.7%, 46.7%]	-10.4% [-28.6%, 8.0%]	0.5165	0.3327

[1]Clopper-Pearson method for estimating exact binomial confidence intervals.

[2]Exact Unconditional Confidence Limits for the Risk Difference.

[3]One-sided non-inferiority test based on Wald method at $\alpha = 0.025$.

[4]Two-Sided Fisher's Exact p-value.

Unavailability of data contributed to fewer subjects analyzed compared to the cohort sample size.

Table 20. CEC-Adjudicated Device-Related Adverse Events Observed in SWIFT Clinical Study

MedDRA Preferred Term	Roll-in (all Solitaire™ FR device)	Randomized Solitaire™ FR device	Randomized Concentric Medical MERCI™ device
Units	% (pts/N) [AEs]	% (pts/N) [AEs]	% (pts/N) [AEs]
Subarachnoid Haemorrhage	3.2% (1/31) [1]	1.7% (1/58) [1]	10.9% (6/55) [6]
Cerebrovascular Spasm	6.5% (2/31) [2]	5.2% (3/58) [3]	5.5% (3/55) [3]
Vasospasm	6.5% (2/31) [2]	3.4% (2/58) [2]	3.6% (2/55) [2]
Carotid Artery Dissection	3.2% (1/31) [1]	1.7% (1/58) [1]	1.8% (1/55) [1]
Cerebral Haematoma	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]
Intraventricular Haemorrhage	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]
Vessel Perforation	0.0% (0/31) [0]	0.0% (0/58) [0]	3.6% (2/55) [2]
Catheter Site Haematoma	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]
Increased Tendency To Bruise	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]
Ischaemic Stroke	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]
Retinal Artery Embolism	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]
Vascular Access Complication	3.2% (1/31) [1]	0.0% (0/58) [0]	0.0% (0/55) [0]
Device Breakage	0.0% (0/31) [0]	1.7% (1/58) [1]	0.0% (0/55) [0]
Haemorrhagic Cerebral Infarction	0.0% (0/31) [0]	1.7% (1/58) [1]	0.0% (0/55) [0]
Hypotension	0.0% (0/31) [0]	1.7% (1/58) [1]	0.0% (0/55) [0]
Procedural Hypertension	0.0% (0/31) [0]	1.7% (1/58) [1]	0.0% (0/55) [0]
Tachycardia	0.0% (0/31) [0]	1.7% (1/58) [1]	0.0% (0/55) [0]
Vascular Pseudoaneurysm	0.0% (0/31) [0]	1.7% (1/58) [1]	0.0% (0/55) [0]
Bradycardia	0.0% (0/31) [0]	0.0% (0/58) [0]	1.8% (1/55) [1]
Cerebral Gas Embolism	0.0% (0/31) [0]	0.0% (0/58) [0]	1.8% (1/55) [1]
Retinal Artery Occlusion	0.0% (0/31) [0]	0.0% (0/58) [0]	1.8% (1/55) [1]
Sinus Bradycardia	0.0% (0/31) [0]	0.0% (0/58) [0]	1.8% (1/55) [1]

Table 21. Screen Failure

Reason for Screen Failure	Occurrence
Angiographic	**91**
No occlusion on angiogram	35
Complete cervical carotid occlusion	28
Occlusion located in ineligible vessel	16
Vessel tortuosity	7
Vessel dissection	5
Clinical	**234**
NIHSS does not qualify for study	81
Age	38
Neurological signs improvement	34
Stroke onset time unknown	29
Time from stroke onset exceed 8 hrs	23
Exclusion criteria not documented	11
Unable to consent	11
No intervention per physician	2
INR exceed required limit	2
Uncontrolled hypertension	1
Subject presented comatose	1
Pre-existing condition, tumor	1
Imaging	**55**
CT/MRI showing infarct	23
Hemorrhage present at baseline	22
Imaging exclusion specific not listed	5
Infarct involving greater than 1/3 of the MCA territory	4
No mismatch on MRI	1
Other	**11**
Total	**391**

SAFETY AND EFFECTIVENESS INFORMATION – INDICATION 3

DEFUSE 3 Clinical Study

The Endovascular Therapy Following Imaging Evaluation for Ischemic Stroke (DEFUSE 3) study was a multicenter, randomized, open-label trial, with blinded outcome assessment, of thrombectomy in patients 6 to 16 hours after they were last known to be well and who had remaining ischemic brain tissue that was not yet infarcted. Patients with proximal middle-cerebral-artery or internal-carotid-artery occlusion, an initial infarct size of less than 70 ml, and a ratio of the volume of ischemic tissue on perfusion imaging to infarct volume of 1.8 or more were randomly assigned to endovascular therapy (thrombectomy) plus standard medical therapy (endovascular-therapy group) or standard medical therapy alone (medical-therapy group). Within the endovascular therapy plus standard medical therapy group, the Solitaire™ endovascular therapy group (Solitaire™ group) was defined as those subjects where the Solitaire™ Revascularization Device was used first amongst other mechanical thrombectomy devices.

Inclusion Criteria

Subjects were considered eligible for the study if they satisfied all of the following inclusion criteria:

- Signs and symptoms consistent with the diagnosis of an acute anterior circulation ischemic stroke
- Age 18-90 years
- Baseline NIHSSS is ≥ 6 and remains ≥6 immediately prior to randomization
- Endovascular treatment can be initiated (vascular puncture) between 6 and 16 hours of stroke onset. Stroke onset is defined as the time the patient was last known to be at their neurologic baseline (wake-up strokes are eligible if they meet the above time limits).
- Modified Rankin Scale less than or equal to 2 prior to qualifying stroke (functionally independent for all ADLs)
- Patient/Legally Authorized Representative has signed the Informed Consent form.

In addition, neuroimaging inclusion criteria included:

- ICA or MCA-M1 occlusion (carotid occlusions can be cervical or intracranial; with or without tandem MCA lesions) by MRA or CTA

AND

- Target Mismatch Profile on CT perfusion or MRI (ischemic core volume is < 70 ml, mismatch ratio is > 1.8 and mismatch volume* is > 15 ml)

***NOTES:** The mismatch volume is determined by the RAPID software in real time based on the difference between the ischemic core lesion volume and the Tmax>6s lesion volume. If both a CT perfusion and a multimodal MRI scan are performed prior to enrollment, the later of the 2 scans is assessed to determine eligibility. Only an intracranial MRA is required for patients screened with MRA; cervical MRA is not required. Cervical and intracranial CTA are typically obtained simultaneously in patients screened with CTA, but only the intracranial CTA is required for enrollment.

Alternative neuroimaging inclusion criteria (if perfusion imaging or CTA/MRA is technically inadequate):

- If CTA (or MRA) is technically inadequate:
  a. Tmax>6s perfusion deficit consistent with an ICA or MCA-M1 occlusion

AND

  b. Target Mismatch Profile (ischemic core volume is < 70 ml, mismatch ratio is >1.8 and mismatch volume is >15 ml as determined by RAPID software)

- If MRP is technically inadequate:
  a. ICA or MCA-M1 occlusion (carotid occlusions can be cervical or intracranial; with or without tandem MCA lesions) by MRA (or CTA, if MRA is technically inadequate and a CTA was performed within 60 minutes prior to the MRI)

AND

  b. DWI lesion volume < 25 ml

- If CTP is technically inadequate:
  a. Patient can be screened with MRI and randomized if neuroimaging criteria are met.

Exclusion Criteria

Subjects were considered ineligible for study participation if they met any of the following exclusion criteria:

- Other serious, advanced, or terminal illness (investigator judgment) or life expectance is less than 6 months.
- Pre-existing medical, neurological or psychiatric disease that would confound the neurological or functional evaluations
- Pregnant
- Unable to undergo a contrast brain perfusion scan with either MRI or CT
- Known allergy to iodine that precludes an endovascular procedure
- Treated with tPA >4.5 hours after time last known well
- Treated with tPA 3-4.5 hours after last known well AND any of the following: age >80, current anticoagulant use, history of diabetes AND prior stroke, NIHSS >25
- Known hereditary or acquired hemorrhagic diathesis, coagulation factor deficiency; recent oral anticoagulant therapy with INR > 3 (recent use of one of the new oral anticoagulants is not an exclusion if estimated GFR > 30 ml/min).
- Seizures at stroke onset if it precludes obtaining an accurate baseline NIHSS
- Baseline blood glucose of <50 mg/dL (2.78 mmol) or >400 mg/dL (22.20 mmol)
- Baseline platelet count < 50,000/uL
- Severe, sustained hypertension (Systolic Blood Pressure >185 mmHg or Diastolic Blood Pressure >110 mmHg)
- Current participation in another investigational drug or device study
- Presumed septic embolus; suspicion of bacterial endocarditis
- Clot retrieval attempted using a neurothrombectomy device prior to 6 hours from symptom onset
- Any other condition that, in the opinion of the investigator, precludes an endovascular procedure or poses a significant hazard to the subject if an endovascular procedure was performed.

In addition, neuroimaging exclusion criteria included:

- ASPECT score <6 on non-contrast CT (if patient is enrolled based on CT perfusion criteria)
- Evidence of intracranial tumor (except small meningioma) acute intracranial hemorrhage, neoplasm, or arteriovenous malformation
- Significant mass effect with midline shift
- Evidence of internal carotid artery dissection that is flow limiting or aortic dissection
- Intracranial stent implanted in the same vascular territory that precludes the safe deployment/removal of the neurothrombectomy device
- Acute symptomatic arterial occlusions in more than one vascular territory confirmed on CTA/MRA (e.g., bilateral MCA occlusions, or an MCA and a basilar artery occlusion).

Study Endpoints

The primary effectiveness endpoint of the study is 90-day blinded evaluation of modified Rankin Scale (mRS). Secondary efficacy endpoints were 1) functional independence at 90 days, defined as an mRS score ≤2 at day 90; and 2) change in NIHSS score between baseline and 24 hours of >8 points or a 24-hour NIHSS of 0-1.

The primary safety outcomes were death within 90 days and the occurrence of symptomatic intracranial hemorrhage within 36 hours, defined as an increase of at least 4 points in the NIHSS score that was associated with brain hemorrhage on imaging within 36 hours after symptom onset.

Results

The DEFUSE 3 study included a total of 182 subjects (92 in the endovascular therapy group and 90 in the control group).

Solitaire™ was the first device used amongst all mechanical thrombectomy devices for 38 subjects in the endovascular therapy group. The intention-to-treat (ITT) Analysis Cohort is thus comprised of 90 in the control group and 38 in the Solitaire™ group.

The DEFUSE 3 study allowed IV t-PA use beyond 3 hours, although IV t-PA is not approved in the United States beyond 3 hours. A total of 11 subjects (5 from the control group, 6 from the Solitaire™ group) were excluded from the primary and secondary efficacy endpoint analyses due to receiving IV t-PA beyond 3 hours and/or carotid stenting. Therefore, the primary and secondary efficacy endpoint analyses consist of 117 subjects (Analysis Cohort - mITT). In the mITT Analysis Cohort, subjects with multiple interventions (n=8) were treated as failures. All analyses presented for the Analysis Cohort (ITT and mITT) are post-hoc analyses. As such, there may be uncertainty in the interpretation of the clinical study results and any statistically meaningful conclusions due to the post-hoc nature of the analyses, limitations in sample size, and lack of pre-specified hypothesis testing.

A summary of the subject disposition is presented below.

Table 22. Summary of Subject Disposition (Analysis Cohort - ITT)

Outcome	Solitaire™ % (n/N)	Control % (n/N)	Overall % (n/N)
Completed study	100.0% (38/38)	97.8% (88/90)	98.4% (126/128)
Completed 90-day visit	89.5% (34/38)	65.7% (65/98)	77.3% (99/128)
Died prior to 90-day visit	10.5% (4/38)	25.6% (23/90)	21.1% (27/128)
Lost to follow-up	0.0% (0/38)	2.2% (2/90)	1.6% (2/128)
Post-discharge	0.0% (0/38)	0.0% (0/90)	0.0% (0/128)
After 30-day visit	0.0% (0/38)	2.2% (2/90)	1.6% (2/128)

A summary of the procedural characteristics are provided below.

Table 23. Procedural Characteristics (Analysis Cohort - ITT)

Outcome	Solitaire™ Mean ± SD (N) [Median] (IQR) or % (n/N)	Control* Mean ± SD (N) [Median] (IQR) or % (n/N)
Transfer Subject	68.4% (26/38)	70.0% (63/90)
IV t-PA Administration	18.4% (7/38)	8.9% (8/90)
General Anesthesia	36.8% (14/38)	-
Onset to randomization (min)	631.2 ± 162.2 (38) [644.5] (474.5,742.0)	653.0 ± 153.5 (90) [644.5] (522.8,780.8)
Onset to arterial puncture (min)	667.9 ± 160.1 (38) [674.5] (527.8,778.0)	-
Time of onset to ED arrival (min)	520.7 ± 162.6 (38) [530.0] (405.5,637.8)	-
ED arrival to arterial puncture (min)	147.2 ± 120.8 (38) [118.5] (80.8,156.8)	-
Imaging to arterial puncture (min)	69.9 ± 31.3 (38) [65.5] (46.0,86.8)	-
Onset to reperfusion (min)	719.0 ± 167.5 (34) [708.0] (569.8,855.2)	-

***Data provided for control group where available**

Standard summary statistics were calculated for all study variables and subject data were analyzed according to the group to which they were randomized. For continuous variables, statistics included means, standard deviations, medians and interquartile ranges. Categorical variables were summarized in frequency distributions.

Analyses were carried out consistent with the intent to treat principle, in that the randomized assignments in DEFUSE 3 were preserved in making group assignments, and subjects in whom Solitaire™ was the first device used constitute the (Analysis Cohort - ITT).Hypothesis testing, including comparison of the primary efficacy endpoint between treatment groups, was conducted using two-sided tests with p-values less than 0.05 deemed significant.

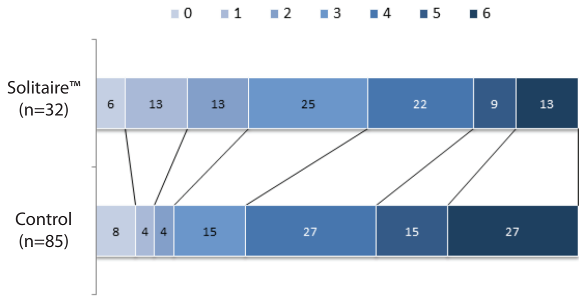
For adverse event reporting, the primary analysis is based on subject counts, not event counts. Both subject counts and event counts are presented in tabular summaries of results.

NOTE: The SFR4-3-20-10, SFR4-3-40-10, SFR4-4-20-05, SFR4-4-20-10, SFR4-4-40-10, SFR4-6-20-10, SFR4-6-24-06 and SFR4-6-40-10 models were not evaluated as part of the DEFUSE 3 study.

Table 24. Primary Effectiveness Endpoint (Analysis Cohort – mITT)

mRS Score at 90 days	Solitaire™ Mean ± SD (N) [Median] (IQR) or % (n/N)	Control Mean ± SD (N) [Median] (IQR) or % (n/N)	p-value
-	3.2 ± 1.7 (32) [3.0] (2.0,4.0)	4.0 ± 1.8 (85) [4.0] (3.0,6.0)	0.014
0	6.2% (2/32)	8.2% (7/85)	-
1	12.5% (4/32)	3.5% (3/85)	-
2	12.5% (4/32)	3.5% (3/85)	-
3	25.0% (8/32)	15.3% (13/85)	-
4	21.9% (7/32)	27.1% (23/85)	-
5	9.4% (3/32)	15.3% (13/85)	-

Table 24. Primary Effectiveness Endpoint (Analysis Cohort – mITT)			
mRS Score at 90 days	**Solitaire™ Mean ± SD (N) [Median] (IQR) or % (n/N)**	**Control Mean ± SD (N) [Median] (IQR) or % (n/N)**	**p-value**
6	12.5% (4/32)	27.1% (23/85)	-



Patients (%)
Figure 5. mRS at 90 days (Analysis Cohort – mITT)

Table 25. Secondary Efficacy Endpoint, mRS 0-2 at 90 days (Analysis Cohort – mITT)		
Outcome	**Solitaire™ % (n/N)**	**Control % (n/N)**
Functional Independence (mRS 0-2) at 90 days	31.2% (10/32)	15.3% (13/85)

Table 26. Early Neurological Improvement at 24h (Analysis Cohort - ITT)		
Outcome	**Solitaire™ % (n/N)7**	**Control % (n/N)**
Early neurological improvement	31.6% (12/38)	5.6% (5/89*)

*NOTE: NIHSS at 24 hours, which is needed to compute early neurological improvement, was available for 89 of 90 subjects in the control group.

Table 27. TICI Immediate Post-Procedure * (Analysis Cohort - mITT)	
Outcome TICI post-procedure (central reading)	**Solitaire™ % (n/N)**
0	21.9% (7/32)
2a	12.5% (4/32)
2b	43.8% (14/32)
3	21.9% (7/32)
TICI≥2b post-procedure	65.6% (21/32)
* TICI was not assessed for Control group immediately post-procedure	

Table 28. Reperfusion and Recanalization 24 hours Post-Procedure (Analysis Cohort - mITT) *		
Outcome	**Solitaire™ Mean ± SD (N) [Median] (IQR) or % (n/N)**	**Control Mean ± SD (N) [Median] (IQR) or % (n/N)**
Reperfusion rate (%) at 24h	92.6 ± 20.2 (24) [100.0] (99.0,100.0)	48.7 ± 46.0 (63) [53.8] (24.7,84.7)
Successful reperfusion (>90%) at 24h	83.3% (20/24)	17.5% (11/63)
Complete recanalization at 24h	82.8% (24/29)	19.2% (14/73)

* Results are presented for subjects in the mITT analysis cohort with available data. Reperfusion percentage and successful reperfusion were available for 24 of 32 subjects in the Solitaire™ group, and 63 of 85 subjects in the control group. Complete recanalization was available for 29 of 32 subjects in the Solitaire™ group and 73 of 85 subjects in the control group.

Table 29. NIHSS through Day 90 (Analysis Cohort - ITT)		
Outcome	**Solitaire™ Mean ± SD (N) [Median] (IQR)**	**Control Mean ± SD (N) [Median] (IQR)**
NIHSS at 24h	12.2 ± 8.7 (38) [9.0] (5.0,19.8)	15.9 ± 7.3 (89*) [16.0] (11.0,21.0)
NIHSS at discharge	10.0 ± 11.4 (37) [5.0] (3.0,13.0)	15.4 ± 10.8 (82) [15.0] (7.0,21.0)
NIHSS at day 30	11.5 ± 14.0 (22) [7.0] (1.3,17.8)	21.8 ± 16.5 (50) [17.0] (9.0,42.0)
NIHSS at day 90	9.8 ± 13.4 (30) [5.0] (2.0,10.8)	19.7 ± 17.3 (67) [14.0] (4.0,42.0)

*NOTE: NIHSS at 24 hours was available for 89 of 90 subjects in the control group.

Table 30. Primary Safety Endpoints (Analysis Cohort - ITT)		
Safety Endpoint	**Solitaire™ % (n/N)**	**Control % (n/N)**
Mortality at 90 days	10.5% (4/38)	25.6% (23/90)
Symptomatic ICH	2.6% (1/38)	4.4% (4/90)

Table 31. All Adverse Events by MedDRA Classification (Analysis Cohort - ITT)			
System Organ Class	**Preferred Term**	**Solitaire™ % (n/N) [events]**	**Control % (n/N) [events]**
TOTAL	**TOTAL**	**78.9% (30/38) [97]**	**86.7% (78/90) [229]**
Blood and lymphatic system disorders	-	**5.3% (2/38) [2]**	**6.7% (6/90) [7]**
-	Anaemia	2.6% (1/38) [1]	4.4% (4/90) [4]
-	Leukocytosis	2.6% (1/38) [1]	3.3% (3/90) [3]
Cardiac disorders	-	**21.1% (8/38) [10]**	**16.7% (15/90) [19]**
-	Atrial fibrillation	10.5% (4/38) [4]	7.8% (7/90) [7]
-	Atrial flutter	-	1.1% (1/90) [1]
-	Atrioventricular block first degree	-	1.1% (1/90) [1]
-	Bradycardia	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Cardiac arrest	-	2.2% (2/90) [2]
-	Cardiac failure chronic	-	1.1% (1/90) [1]
-	Cardiac failure congestive	2.6% (1/38) [1]	-
-	Cardiac ventricular thrombosis	2.6% (1/38) [1]	-
-	Cardio-respiratory arrest	5.3% (2/38) [2]	1.1% (1/90) [1]
-	Left ventricular dysfunction	-	1.1% (1/90) [1]
-	Sinus bradycardia	2.6% (1/38) [1]	2.2% (2/90) [2]
-	Tachycardia	-	2.2% (2/90) [2]
Gastrointestinal disorders	-	**21.1% (8/38) [11]**	**12.2% (11/90) [11]**
-	Abdominal pain upper	-	1.1% (1/90) [1]
-	Aphthous ulcer	2.6% (1/38) [1]	-
-	Bowel movement irregularity	-	1.1% (1/90) [1]
-	Constipation	-	1.1% (1/90) [1]
-	Diarrhoea	-	1.1% (1/90) [1]
-	Dysphagia	5.3% (2/38) [2]	4.4% (4/90) [4]

-	Gastrointestinal haemorrhage	-	1.1% (1/90) [1]
-	Lower gastrointestinal haemorrhage	2.6% (1/38) [1]	-
-	Nausea	5.3% (2/38) [2]	1.1% (1/90) [1]
-	Pancreatitis acute	2.6% (1/38) [1]	-
-	Retroperitoneal haematoma	2.6% (1/38) [1]	-
-	Upper gastrointestinal haemorrhage	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Vomiting	5.3% (2/38) [2]	-
General disorders and administration site conditions	-	**13.2% (5/38) [5]**	**12.2% (11/90) [12]**
-	Asthenia	2.6% (1/38) [1]	-
-	Chills	-	1.1% (1/90) [1]
-	Death	-	1.1% (1/90) [1]
-	Drug withdrawal syndrome	-	1.1% (1/90) [1]
-	Infusion site extravasation	-	1.1% (1/90) [1]
-	Oedema peripheral	-	1.1% (1/90) [1]
-	Pain	-	2.2% (2/90) [2]
-	Pyrexia	10.5% (4/38) [4]	4.4% (4/90) [4]
-	Tenderness	-	1.1% (1/90) [1]
Infections and infestations	-	**10.5% (4/38) [4]**	**24.4% (22/90) [28]**
-	Bacteraemia	2.6% (1/38) [1]	-
-	Bronchitis	-	1.1% (1/90) [1]
-	Cellulitis	-	1.1% (1/90) [1]
-	Clostridium difficile colitis	-	1.1% (1/90) [1]
-	Clostridium difficile infection	-	1.1% (1/90) [1]
-	Conjunctivitis	-	1.1% (1/90) [1]
-	Fungal infection	-	1.1% (1/90) [1]
-	Pneumonia	2.6% (1/38) [1]	4.4% (4/90) [4]
-	Pneumonia klebsiella	-	1.1% (1/90) [1]
-	Sepsis	2.6% (1/38) [1]	4.4% (4/90) [4]
-	Septic shock	-	1.1% (1/90) [1]
-	Urinary tract infection	2.6% (1/38) [1]	11.1% (10/90) [10]
-	Urinary tract infection bacterial	-	1.1% (1/90) [1]
-	Urosepsis	-	1.1% (1/90) [1]
Injury, poisoning and procedural complications	-	**7.9% (3/38) [3]**	**4.4% (4/90) [4]**
-	Fall	-	2.2% (2/90) [2]
-	Feeding tube complication	-	1.1% (1/90) [1]
-	Incision site pain	2.6% (1/38) [1]	-
-	Laceration	-	1.1% (1/90) [1]
-	Rib fracture	2.6% (1/38) [1]	-
-	Vascular pseudoaneurysm	2.6% (1/38) [1]	-
Investigations	-	**5.3% (2/38) [2]**	**5.6% (5/90) [5]**
-	Blood creatinine increased	2.6% (1/38) [1]	-
-	Blood glucose increased	-	1.1% (1/90) [1]
-	Ejection fraction decreased	-	1.1% (1/90) [1]
-	Transaminases increased	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Troponin	-	1.1% (1/90) [1]
-	Troponin increased	-	1.1% (1/90) [1]
Metabolism and nutrition disorders	-	**13.2% (5/38) [7]**	**14.4% (13/90) [17]**
-	Diabetes mellitus	2.6% (1/38) [1]	2.2% (2/90) [2]
-	Gout	2.6% (1/38) [1]	-
-	Hyperglycaemia	-	4.4% (4/90) [4]
-	Hyperlipidaemia	-	1.1% (1/90) [1]
-	Hypernatraemia	-	1.1% (1/90) [1]
-	Hypoglycaemia	2.6% (1/38) [1]	-
-	Hypokalaemia	5.3% (2/38) [2]	5.6% (5/90) [5]
-	Hyponatraemia	2.6% (1/38) [1]	-
-	Hypophosphataemia	2.6% (1/38) [1]	2.2% (2/90) [2]
-	Malnutrition	-	2.2% (2/90) [2]
Musculoskeletal and connective tissue disorders	-	**7.9% (3/38) [3]**	**4.4% (4/90) [5]**
-	Back pain	2.6% (1/38) [1]	3.3% (3/90) [3]
-	Musculoskeletal pain	2.6% (1/38) [1]	-
-	Neck pain	-	1.1% (1/90) [1]
-	Osteoarthritis	-	1.1% (1/90) [1]
-	Pain in jaw	2.6% (1/38) [1]	-
Nervous system disorders	-	**42.1% (16/38) [22]**	**58.9% (53/90) [64]**
-	Aphasia	-	1.1% (1/90) [1]
-	Brain oedema	5.3% (2/38) [3]	3.3% (3/90) [3]
-	Carotid artery dissection	2.6% (1/38) [1]	-
-	Cerebellar haemorrhage	2.6% (1/38) [1]	-
-	Cerebral haemorrhage	-	6.7% (6/90) [6]
-	Cerebral infarction	-	4.4% (4/90) [5]
-	Cerebrovascular accident	2.6% (1/38) [1]	5.6% (5/90) [5]
-	Encephalopathy	-	1.1% (1/90) [1]
-	Haemorrhage intracranial	2.6% (1/38) [1]	3.3% (3/90) [3]
-	Haemorrhagic transformation stroke	23.7% (9/38) [9]	14.4% (13/90) [13]
-	Headache	-	4.4% (4/90) [4]
-	Intracranial aneurysm	2.6% (1/38) [1]	-
-	Metabolic encephalopathy	2.6% (1/38) [1]	-
-	Neurological decompensation	2.6% (1/38) [1]	8.9% (8/90) [8]
-	Seizure	-	2.2% (2/90) [2]
-	Sinus headache	-	1.1% (1/90) [1]
-	Somnolence	2.6% (1/38) [1]	-
-	Spinal cord oedema	2.6% (1/38) [1]	-
-	Stroke in evolution	-	13.3% (12/90) [12]
-	Subarachnoid haemorrhage	2.6% (1/38) [1]	-
Psychiatric disorders	-	**13.2% (5/38) [5]**	**4.4% (4/90) [4]**
-	Agitation	-	3.3% (3/90) [3]
-	Alcohol withdrawal syndrome	2.6% (1/38) [1]	-
-	Anxiety	2.6% (1/38) [1]	-
-	Depression	5.3% (2/38) [2]	-
-	Insomnia	2.6% (1/38) [1]	1.1% (1/90) [1]
Renal and urinary disorders	-	**10.5% (4/38) [4]**	**4.4% (4/90) [5]**
-	Acute kidney injury	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Chronic kidney disease	2.6% (1/38) [1]	-
-	Urinary incontinence	-	2.2% (2/90) [2]
-	Urinary retention	5.3% (2/38) [2]	2.2% (2/90) [2]
Respiratory, thoracic and mediastinal disorders	-	**13.2% (5/38) [5]**	**25.6% (23/90) [29]**
-	Acute respiratory failure	-	2.2% (2/90) [2]
-	Atelectasis	-	2.2% (2/90) [2]
-	Chronic obstructive pulmonary disease	2.6% (1/38) [1]	1.1% (1/90) [1]

-	Dyspnoea	-	1.1% (1/90) [1]
-	Epistaxis	-	1.1% (1/90) [1]
-	Hypoxia	-	1.1% (1/90) [1]
-	Laryngeal oedema	-	1.1% (1/90) [1]
-	Pleural effusion	-	1.1% (1/90) [1]
-	Pneumonia aspiration	2.6% (1/38) [1]	11.1% (10/90) [10]
-	Pneumothorax	-	2.2% (2/90) [2]
-	Pulmonary embolism	2.6% (1/38) [1]	3.3% (3/90) [3]
-	Respiratory arrest	2.6% (1/38) [1]	-
-	Respiratory failure	2.6% (1/38) [1]	2.2% (2/90) [2]
-	Tachypnoea	-	1.1% (1/90) [1]
-	Upper respiratory tract congestion	-	1.1% (1/90) [1]
Skin and subcutaneous tissue disorders	**-**	**2.6% (1/38) [1]**	**-**
-	Rash	2.6% (1/38) [1]	-
Vascular disorders	**-**	**26.3% (10/38) [13]**	**16.7% (15/90) [16]**
-	Deep vein thrombosis	5.3% (2/38) [2]	3.3% (3/90) [3]
-	Haematoma	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Hypertension	7.9% (3/38) [3]	4.4% (4/90) [4]
-	Hypotension	13.2% (5/38) [5]	6.7% (6/90) [6]
-	Thrombophlebitis superficial	-	1.1% (1/90) [1]
-	Thrombosis	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Vessel perforation	2.6% (1/38) [1]	-
Congenital, familial and genetic disorders	**-**	**-**	**1.1% (1/90) [1]**
-	Atrial septal defect	-	1.1% (1/90) [1]
Eye disorders	**-**	**-**	**1.1% (1/90) [1]**
-	Visual acuity reduced	-	1.1% (1/90) [1]
Product issues	-	-	1.1% (1/90) [1]
-	**Device occlusion**	-	**1.1% (1/90) [1]**

"-" indicates not reported in Solitaire™ or Control group

Table 32. Serious Adverse Events by MedDRA Classification (Analysis Cohort - ITT)

System Organ Class	Preferred Term	Solitaire™ % (n/N) [events]	Control % (n/N) [events]
TOTAL	**TOTAL**	**47.4% (18/38) [26]**	**53.3% (48/90) [68]**
Cardiac disorders	**-**	**5.3% (2/38) [2]**	**4.4% (4/90) [5]**
-	Atrial fibrillation	-	2.2% (2/90) [2]
-	Cardiac arrest	-	2.2% (2/90) [2]
-	Cardio-respiratory arrest	5.3% (2/38) [2]	1.1% (1/90) [1]
Gastrointestinal disorders	**-**	**7.9% (3/38) [4]**	**3.3% (3/90) [3]**
-	Constipation	-	1.1% (1/90) [1]
-	Gastrointestinal haemorrhage	-	1.1% (1/90) [1]
-	Lower gastrointestinal haemorrhage	2.6% (1/38) [1]	-
-	Pancreatitis acute	2.6% (1/38) [1]	-
-	Retroperitoneal haematoma	2.6% (1/38) [1]	-
-	Upper gastrointestinal haemorrhage	2.6% (1/38) [1]	1.1% (1/90) [1]
General disorders and administration site conditions	**-**	**2.6% (1/38) [1]**	**1.1% (1/90) [1]**
-	Asthenia	2.6% (1/38) [1]	-
-	Death	-	1.1% (1/90) [1]
Infections and infestations	**-**	**2.6% (1/38) [1]**	**7.8% (7/90) [8]**
-	Pneumonia	-	1.1% (1/90) [1]
-	Pneumonia klebsiella	-	1.1% (1/90) [1]
-	Sepsis	2.6% (1/38) [1]	4.4% (4/90) [4]
-	Septic shock	-	1.1% (1/90) [1]
-	Urosepsis	-	1.1% (1/90) [1]
Nervous system disorders	**-**	**23.7% (9/38) [10]**	**31.1% (28/90) [30]**
-	Aphasia	-	1.1% (1/90) [1]
-	Brain oedema	5.3% (2/38) [2]	3.3% (3/90) [3]
-	Carotid artery dissection	2.6% (1/38) [1]	-
-	Cerebral haemorrhage	-	2.2% (2/90) [2]
-	Cerebral infarction	-	1.1% (1/90) [1]
-	Cerebrovascular accident	2.6% (1/38) [1]	5.6% (5/90) [5]
-	Encephalopathy	-	1.1% (1/90) [1]
-	Haemorrhage intracranial	2.6% (1/38) [1]	2.2% (2/90) [2]
-	Haemorrhagic transformation stroke	2.6% (1/38) [1]	3.3% (3/90) [3]
-	Metabolic encephalopathy	2.6% (1/38) [1]	-
-	Neurological decompensation	2.6% (1/38) [1]	3.3% (3/90) [3]
-	Somnolence	2.6% (1/38) [1]	-
-	Stroke in evolution	-	10.0% (9/90) [9]
-	Subarachnoid haemorrhage	2.6% (1/38) [1]	-
Psychiatric disorders	**-**	**2.6% (1/38) [1]**	**-**
-	Depression	2.6% (1/38) [1]	-
Renal and urinary disorders	**-**	**5.3% (2/38) [2]**	**-**
-	Acute kidney injury	2.6% (1/38) [1]	-
-	Chronic kidney disease	2.6% (1/38) [1]	-
Respiratory, thoracic and mediastinal disorders	**-**	**10.5% (4/38) [4]**	**14.4% (13/90) [15]**
-	Acute respiratory failure	-	2.2% (2/90) [2]
-	Atelectasis	-	1.1% (1/90) [1]
-	Pneumonia aspiration	2.6% (1/38) [1]	5.6% (5/90) [5]
-	Pneumothorax	-	2.2% (2/90) [2]
-	Pulmonary embolism	2.6% (1/38) [1]	3.3% (3/90) [3]
-	Respiratory arrest	2.6% (1/38) [1]	-
-	Respiratory failure	2.6% (1/38) [1]	2.2% (2/90) [2]
Vascular disorders	**-**	**2.6% (1/38) [1]**	**3.3% (3/90) [3]**
-	Haematoma	-	1.1% (1/90) [1]
-	Hypotension	2.6% (1/38) [1]	1.1% (1/90) [1]
-	Thrombosis	-	1.1% (1/90) [1]
Injury, poisoning and procedural complications	**-**	**-**	**2.2% (2/90) [2]**
-	Fall	-	1.1% (1/90) [1]
-	Feeding tube complication	-	1.1% (1/90) [1]
Product issues	**-**	**-**	**1.1% (1/90) [1]**
-	Device occlusion	-	1.1% (1/90) [1]

"-" indicates not reported in Solitaire™ or Control group

Table 33. Procedure Related Adverse Events by MedDRA Classification (Analysis Cohort - ITT)

System Organ Class	Preferred Term	Solitaire™ % (n/N) [events]	Control % (n/N) [events]
TOTAL	**TOTAL**	**28.9% (11/38) [14]**	**4.4% (4/90) [4]**
Cardiac disorders	**-**	**2.6% (1/38) [1]**	**1.1% (1/90) [1]**

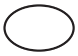
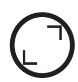
Table 33. Procedure Related Adverse Events by MedDRA Classification (Analysis Cohort - ITT)			
System Organ Class	**Preferred Term**	**Solitaire™ % (n/N) [events]**	**Control % (n/N) [events]**
-	Atrial fibrillation	-	1.1% (1/90) [1]
-	Bradycardia	2.6% (1/38) [1]	-
Injury, poisoning and procedural complications	**-**	**2.6% (1/38) [1]**	**-**
-	Vascular pseudoaneurysm	2.6% (1/38) [1]	-
Nervous system disorders	**-**	**23.7% (9/38) [10]**	**1.1% (1/90) [1]**
-	Brain oedema	2.6% (1/38) [1]	-
-	Cerebellar haemorrhage	2.6% (1/38) [1]	-
-	Cerebral haemorrhage	-	1.1% (1/90) [1]
-	Cerebrovascular accident	2.6% (1/38) [1]	-
-	Haemorrhagic transformation stroke	18.4% (7/38) [7]	-
Vascular disorders	**-**	**5.3% (2/38) [2]**	**-**
-	Haematoma	2.6% (1/38) [1]	-
-	Vessel perforation	2.6% (1/38) [1]	-
General disorders and administration site conditions	**-**	**-**	**1.1% (1/90) [1]**
-	Infusion site extravasation	-	1.1% (1/90) [1]
Investigations	**-**	**-**	**1.1% (1/90) [1]**
-	Troponin increased	-	1.1% (1/90) [1]

"-" indicates not reported in Solitaire™ or Control group

WARRANTY DISCLAIMER

Although this product has been manufactured under carefully controlled conditions, the manufacturer has no control over the conditions under which this product is used. The manufacturer therefore disclaims all warranties, both expressed and implied, with respect to the product including, but not limited to, any implied warranty of merchantability or fitness for a particular purpose. The manufacturer shall not be liable to any person or entity for any medical expenses or any direct, incidental or consequential damages caused by any use, defect, failure or malfunction of the product, whether a claim for such damages is based upon warranty, contract, tort or otherwise. No person has any authority to bind the manufacturer to any representation or warranty with respect to the product. The exclusions and limitation set out above are not intended to, and should not be construed so as to contravene mandatory provisions of applicable law. If any part or term of this Disclaimer of Warranty is held illegal, unenforceable or in conflict with applicable law by a court of competent jurisdiction, the validity of the remaining portions of this Disclaimer of Warranty shall not be affected, and all rights and obligations shall be construed and enforced as if this Disclaimer of Warranty did not contain the particular part or term held to be invalid.

en Symbol Glossary

STERILE EO	**en** Sterilized using ethylene oxide		**en** Keep dry
	**en** Single sterile barrier system	REF	**en** Catalogue number
	**en** Do not re-use		**en** Manufacturer
Rx ONLY	**en** Caution: Federal (USA) law restricts this device to sale by or on the order of a physician		**en** Use-by date
	**en** Do not resterilize	LOT	**en** Batch code
 www.medtronic.com/manuals	**en** Consult electronic instructions for use		**en** Date of manufacture
	**en** Caution	CONTENTS	**en** Contents of Package
	**en** Do not use if package is damaged and consult instructions for use	MD	**en** Medical device
	**en** Non-pyrogenic		**en** Inner diameter (compatible microcatheter)
	**en** Keep away from sunlight		

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